Introduction

A laptop computer resembles the human brain in volume and power use—but it is stupid. Deep Blue, the IBM supercomputer that crushed Grandmaster Garry Kasparov at chess, is 100,000 times larger and draws 100,000 times more power (figure I.1). Yet, despite Deep Blue's excellence at chess, it too is stupid, the electronic equivalent of an idiot savant. The computer operates at the speed of light whereas the brain is slow. So, wherein lies the brain's advantage? A short answer is that the brain employs a hybrid architecture of superior design. A longer answer is this book—whose purpose is to identify the sources of such computational efficiency.

The brain's inner workings have been studied scientifically for more than a century—initially by a few investigators with simple methods. In the last 20 years the field has exploded, with roughly 50,000 neuroscientists applying increasingly advanced methods. This outburst amounts to 1 million person-years of research—and facts have accumulated like a mountain. At the base are detailed descriptions: of neural connections and electrical responses, of functional images that correlate with mental states, and of molecules such as ion channels, receptors, G proteins, and so on. Higher up are key discoveries about mechanism: the action potential, transmitter release, synaptic excitation and inhibition. Summarizing this Everest of facts and mechanisms, there exist superb compendia (Kandel et al., 2012; Purves et al., 2012; Squire et al., 2008).

But what if one seeks a book to set out principles that explain how our brain, while being far smarter than a supercomputer, can also be far smaller and cheaper? Then the shelf is bare. One reason is that modern neuroscience has been "technique driven." Whereas in the 1960s most experiments that one might conceive were technically impossible, now with methods such as patch clamping, two-photon microscopy, and functional magnetic resonance imaging (fMRI), aided by molecular biology, the situation has reversed, and it is harder to conceive of an experiment that can*not* be done.

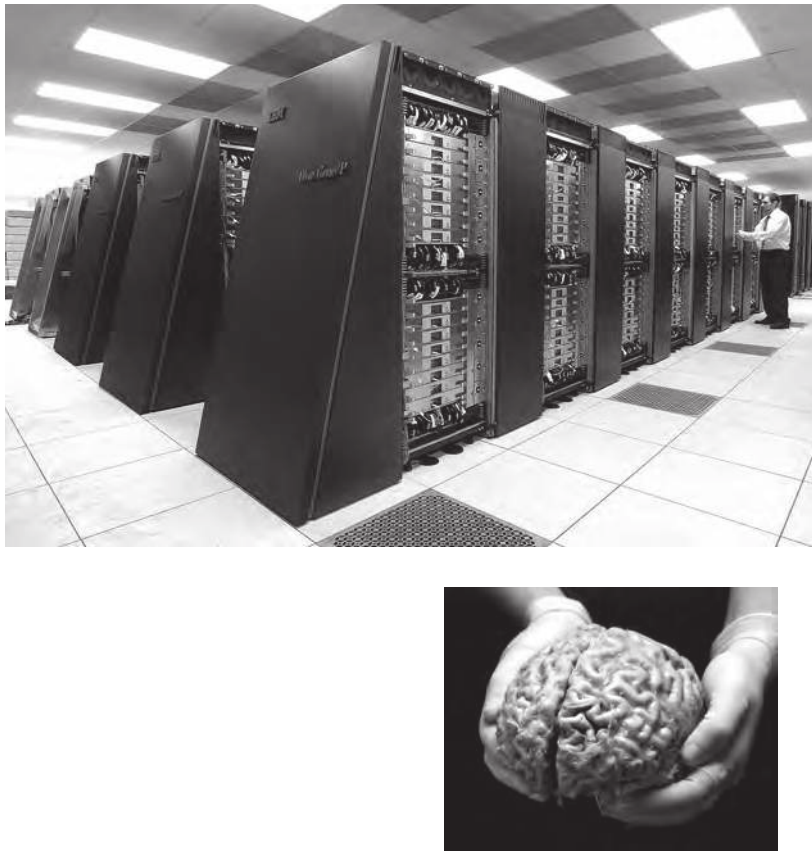


Figure I.1
How do neural circuits use space and power so efficiently? Computer: Image http://upload.wikimedia.org/wikipedia/commons/d/d3/IBM_Blue_Gene_P_supercomputer.jpg. Brain: Photo by UW-Madison, University Communications © Board of Regents of the University of Wisconsin System.

Consequently, the idea of pausing to distill principles from facts has lacked appeal. Moreover, to many who ferret out great new facts for a living, it has seemed like a waste of time.

Yet, we draw inspiration from Charles Darwin, who remarked, "My mind seems to have become a kind of machine for grinding general laws out of large collections of facts" (Darwin, 1881). Darwin, of course, is incomparable, but this is sort of how our minds work too. So we have written a small book—relative to the great compendia—intending to beat a rough path up "Data Mountain" in search of organizing principles.

Principles of engineering

The brain is a physical device that performs specific functions; therefore, its design must obey general principles of engineering. Chapter 1 identifies several that we have gleaned (not being engineers) from essays and books on mechanical and electrical design. These principles do not address specific questions about the brain, but they do set a context for ordering one's thoughts—especially helpful for a topic so potentially intimidating. For example, it helps to realize that neuroscience is really an exercise in "reverse engineering"—disassembling a device in order to understand it.

This insight points immediately to a standard set of questions that we suppose are a mantra for all "reverse engineers": *What does it do? What are its specifications? What is the environmental context?* Then there are commandments, such as *Study the interfaces* and *Complicate the design*. The latter may puzzle scientists who, in explaining phenomena, customarily strive for simplicity. But engineers focus on designing effective devices, so they have good reasons to complicate.[1] This commandment, we shall see, certainly applies to the brain.

Why a brain?

To address the engineer's first question, we consider why an animal should need a brain—what fundamental purpose does it serve and at what cost to the organism? Chapter 2 begins with a tiny bacterium, *Escherichia coli* which succeeds *without* a brain, in order to evaluate what the bacterium can do and what it cannot. Then on to a protozoan, *Paramecium caudatum*, still a single cell and brainless, but so vastly larger than *E. coli* (300,000-fold) that it requires a faster type of signaling. This prefigures long-distance signaling by neurons in multicellular organisms.

The chapter closes with the tiny nematode worm, *Caenorhabditis elegans*, which does have a brain—with exactly 302 neurons. This number is small in absolute terms, but it represents nearly one third of the creature's total cells, so it is a major investment that better turn a profit, and it does. For example, it controls a multicellular system that finds, ingests, and digests bacteria and that allows the worm to recall for several hours the locations of favorable temperatures and bacterial concentrations.

Humans naturally tend to discount the computational abilities of small organisms—which seem, well . . ., mentally deficient—nearly devoid of learning or memory. But small organisms *do* learn and remember. It's just

that their memories match their life contexts: they remember only what they need to and for just long enough. Furthermore, the mechanisms that they evolved for these computations are retained in our own neurons—so we shall see them again.

The progression bacterium → protozoan → worm is accompanied by increasing computational complexity. It is rewarded by increasing capacity to inhabit richer environments and thus to move up the food chain: protozoa eat bacteria, and worms eat protozoa. As engineering, this makes perfect sense: little beasts compute only what they must; thus they pay only for what they use. This is equally true for beasts with much larger brains discussed in chapter 3.

Why a bigger brain?

The brain of a fruit fly (*Drosophila melanogaster*) is 350-fold larger than *C. elegans*', and the brain of a human (*Homo sapiens*) is a million-fold larger than the fly's. These larger brains emerge from the same process of natural selection as the smaller ones, so we should continue to expect from them nothing superfluous—only mechanisms that are essential and pay for themselves. We should also expect that when a feature works really well, it will be retained—like the wheel, the paper clip, the aluminum beer can, and the transistor (Petroski, 1996; Arthur, 2009). We note design features that brains have conserved (with suitable elaborations) across at least 400 million years of natural selection. These features in the human brain are often described as "primitive"—reptilian—reflecting what are considered negative aspects of our nature. But, of course, any feature that has been retained for so long must be pretty effective.

This chapter identifies the core task of all brains: it is to regulate the organism's internal milieu—by responding to needs and, better still, by anticipating needs and preparing to satisfy them before they arise. The advantages of omniscience encourage omnipresence. Brains tend to become universal devices that tune all internal parameters to improve overall stability and economy. "Anticipatory regulation" replaces the more familiar "homeostatic regulation"—which is supposed to operate by waiting for each parameter to deviate from a "set point," then detecting the error and correcting it by feedback. Most physiological investigations during the 20th century were based on the homeostatic model—how kidney, gut, liver, pancreas, and so on work independently, despite Pavlov's early demonstration of the brain's role in anticipatory regulation (Pavlov, 1904). But gradually anticipatory control has been recognized.

Anticipatory regulation offers huge advantages.[2] First, it matches overall response capacity to fluctuations in demand—there should always be enough but not too much. Second, it matches capacity at each stage in the system to anticipated needs downstream, thus threading an efficient path between excess capacity (costly storage) and failure from lack of supplies. Third, it resolves potential conflict between organs by setting and shifting priorities. For example, during digestion it can route more blood to the gut and less to muscle and skin, and during exercise it can reverse this priority. This allows the organism to operate with a smaller blood volume than would otherwise be needed. Finally, it minimizes errors—which are potentially lethal and also cause cumulative damage.

Anticipatory regulation includes behavior

An organ that anticipates need and regulates the internal milieu by overarching control of physiology would be especially effective if it also regulated behavior. For example, it could reduce a body's need for physiological cooling (e.g., sweating—which costs energy and resources—sodium and water) by directing an animal to find shade. Moreover, it could evoke the memory of an unpleasant heatstroke to remind the animal to take anticipatory measures (travel at night, carry water). Such anticipatory mechanisms are driven ceaselessly by *memories* of hunger, cold, drought, or predation: *Pick the beans! Chop wood! Build a reservoir! Lock the door!*

The memories of danger and bad times that shape our behavior can be our own, but often they are stored in the brains of our parents and grandparents. We are reared with *their* nightmares—the flood, the drought, the famine, the pogrom. Before written history, which spans only 6,000 years, all lessons that would help one anticipate and thus avoid a lethal situation could be transmitted only by oral tradition—the memory of a human life span. Given that the retention of memories in small brains corresponds to their useful span, and that retention has a cost, human memory for great events should remain vivid with age whereas recent memories of lesser events should fade (chapter 14).

The most persistent dangers and opportunities, those extending far beyond a few generations, eventually become part of the neural wiring. Monkeys universally fear snakes, and so do most humans—suggesting that the response was encoded into brain structure before the lines split—on the order of 35 million years. But beyond alertness for predators, primate societies reserve their most acute observations and recall for relationships within the family and the troop. The benefit is that an individual's chances

for survival and reproduction are enhanced by the *group's* ability to anticipate and regulate. The cost is that the individual must continuously sense the social structure—in its historical context—to receive aid when needed and to avoid being killed or cast out (Cheney & Seyfarth, 2007).

Consequently, primate brains have expanded significantly in parts concerned with social recognition and planning—such as prefrontal cortex and amygdala. Humans greatly expand these areas and also those for social communication, such as for language, facial expression, and music. These regions serve both the cooperative and the competitive aspects of anticipatory regulation to an awesome degree. They account for much of our brain structure and many of our difficulties.

Flies too show anticipatory behavior—to a level consonant with their life span and environmental reach. A fly need not wait for its blood sugar to fall dangerously low, nor for its temperature to soar dangerously high, before taking action. Instead its brain expresses prewired commands: *Find fruit! In a cool spot!* Anticipatory commands are often tuned to environmental regularities that predict when and where a resource is most likely to appear—or disappear. Thus, circadian rhythms govern foraging and sleep. Seasonal rhythms, which broadly affect resource availability, govern mating and reproduction. Consequently, specific brain hormones tuned to day length send orders to prewired circuits: *Court a mate! Intimidate a competitor!*

What drives behavior?

To ensure that an organism will execute these orders, there are neural mechanisms to make it "feel bad" when a job is undone and "feel good" when it has succeeded. These are circuits whose activity humans experience, respectively, as "anxiety" and "pleasure." Of course, we cannot know what worms or flies experience—but the same neurochemicals drive similar behaviors. This is one wheel that has certainly been decorated over hundreds of millions of years, but not reinvented.

To actually accomplish a task is vastly complicated. Reconsider Deep Blue's task. Each side in chess has 16 pieces—that move one at a time, slowly (minutes), and only in two dimensions. Each piece is constrained to move only in certain ways, and some pieces repeat so that each side has only six different types of motion. This relatively simple setup generates so many possible moves that to evaluate them requires a Deep Blue.

But the organ responsible for anticipatory regulation takes continuous data from every sensory neuron in the organism—both internal and

external—plus myriad hormones and other chemicals. While doing so, it is calculating in real time—milliseconds—how to adjust every body component inside and out. It is flying the fly, finding its food, shade, and mate; it is avoiding predators and intimidating competitors—all the while tweaking every *internal* parameter to match what is about to be needed. Thus, it seems fair to say that Deep Blue is stupid even compared to a fruit fly. This defines sharply the next engineering question: what constrains the design of an effective and efficient brain?

What constrains neural design?

When Hillel was asked in the first century B.C.E. to explain the whole Torah while standing on one leg, he was ready: "That which is hateful to you, do not unto another. The rest is commentary—and now go study."

There is a one-leg answer for neural design: "As information rate rises, costs rise disproportionately." For example, to transmit more information by spikes requires a higher spike rate. Axon diameter rises linearly with spike rate, but axon volume and energy consumption rise as the diameter squared. Thus, the essence of neural design: "Send only information that is needed, and send it as slowly as possible" (chapter 3). This key injunction profoundly shapes the brain's macroscopic layout, as explained in chapter 4. We hope that readers will . . . go study.

If spikes were energetically cheap, their rates would matter less. However, a 100-mV spike requires far more current than a 1-mV response evoked by one packet of chemical transmitter. Obviously then, it is cheaper to compute with the smaller currents. This exemplifies another design principle: minimize energy per *bit* of information by computing at the finest possible level. Chapter 5 identifies this level as a change in protein folding on the scale of nanometers. Such a change can capture, store, and transmit one bit at an energetic cost that approaches the thermodynamic limit. Chapter 6 explains how proteins couple to form intracellular circuits on the scale of micrometers, and chapter 7 explains how a neuron assembles such circuits into devices on a scale of micrometers to millimeters.

It emerges that to compute most efficiently in space and energy, neural circuits should *nanofy*:

1. Make each component irreducibly small: a functional unit should be a single protein molecule (a channel), or a linear polymer of protein subunits (a microtubule), or a sandwich of monomolecular layers (a membrane).

2. Combine irreducible components: a membrane to separate charge and thus permit a voltage, a protein transporter to pump ions selectively across the membrane and actually separate the charges (charge the battery), a pore for ions to flow singly across the membrane and thus create a current, a "gate" to start and stop a current, an amplifier to enlarge the current, and an adaptive mechanism to match a current to circumstance.
3. Compute with *chemistry* wherever possible: regulate gates, amplifiers, and adaptive mechanisms by binding/unbinding small molecules that are present in sufficient numbers to obey the laws of mass action. Achieve speed with chemistry by keeping the volumes small.
4. For speed over distance compute *electrically*: convert a signal computed by chemistry to a current that charges membrane capacitance to spread passively up to a millimeter. For longer distance, regenerate the current by appropriately clustered voltage-gated channels.

Design in the visual system

Having discussed protein computing and miniaturization as general routes to efficiency, we exemplify these points in an integrated system—phototransduction (chapter 8). The engineering challenge is to capture light reflected from objects in the environment in order to extract informative patterns to guide behavior. Transduction employs a biochemical cascade with about half a dozen stages to amplify the energy of individual photons by up to a million-fold while preserving the information embodied as signal-to-noise ratio (S/N) and bandwidth. We explain why so many stages are required.

The photoreceptor signal, once encoded as a graded membrane voltage, spreads passively down the axon to the synaptic terminal. There the analogue signal is digitized as a stream of synaptic vesicles. The insect brain can directly read out this message with very high efficiency because the distance is short enough for passive signaling (chapter 9). The mammal brain can*not* directly read out this vesicle stream because the distance is too great for passive signaling. The mammal eye must transmit by action potentials, but the photoreceptor's analogue signal contains more information than action potentials can encode. Therefore, on-site retinal processing is required (chapters 10, 11).

Principles at higher levels

The principles of neural design at finer scales and lower levels also apply at larger scales and higher levels. For example, they can explain why the first

visual area (V1) in cerebral cortex enormously expands the number and diversity of neurons. And why diverse types project in parallel from V1 to other cortical areas. And why cortex uses many specific areas and arranges them in a particular way. The answers, as explained in chapter 12, are always the same: diverse circuits allow the brain to send only information that is needed and to send it at lower information rates. This holds computation to the steep part of the benefit/cost curve.

Wiring efficiency

Silicon circuits with very large-scale integration strive for optimal layout—to achieve best performance for least space, time, and energy. Neural circuits do the same and thereby produce tremendous diversity of neuronal structure at all spatial scales. For example, cerebellar output neurons (*Purkinje cells*) use a two-dimensional dendritic arbor whereas cerebral output neurons (*pyramidal cells*) use a three-dimensional arbor. Both circuits employ a layered architecture, but the large Purkinje neurons lie *above* a layer of tiny neurons whereas the large pyramidal neurons lie *below* the smaller neurons. Cerebellar cortex folds intensely on a millimeter scale whereas cerebral cortex on this scale is smooth.

Such differences originate from a ubiquitous biophysical constraint: the irreducible electrical resistance of neuronal cytoplasm. Passive signals spread spatially and temporally only as the square root of dendritic diameter ($\sqrt{d}$). This causes a second law of diminishing returns: a dendrite, to double its conduction distance or halve its conduction delay, must quadruple its volume. This prevents neural wires from being any finer and prevents local circuits from being any more voluminous. In both cases conduction delays would grow too large. The constraint on volume drives efficient layout: equal lengths of dendrite and axon and an optimum proportion of wire and synapses. Chapter 13 will explain.

Designs for learning

All organisms use new information to better anticipate the future. Thus, learning is a deep principle of biological design, and therefore of neural design. Accordingly, the brain continually updates its knowledge of every internal and external parameter—which means that learning is also a brain function. As such, neural learning is subject to the same constraints as all other neural functions. It is a design principle that must obey all the others.

To conserve space, time, and energy, new information should be stored at the site where it is processed and from whence it can be recalled without

further expense. This is the synapse. Low-level synapses relay short-term changes in input, so their memories should be short, like that of a bacterium or worm. These synapses should encode at the cheapest levels, by modifying the structure and distribution of proteins. High-level synapses encode conclusions after many stages of processing, so their memories deserve to be longer and encoded more stably, by enlarging the synapse and adding new ones.

A new synapse of diameter (d) occupies area on the postsynaptic membrane as d^2 and volume as d^3. Because adding synapses increases costs disproportionately, learning in an adult brain of fixed volume is subject to powerful space constraints. For every synapse enlarged or added, another must be shrunk or removed. Design of learning must include the principle "save only what is needed." Chapter 14 explains how this plays out in the overall design.

Design and designer

This book proposes that many aspects of the brain's design can be understood as adaptations to improve efficiency under resource constraints. Improvements to brain efficiency must certainly improve fitness. Darwin himself noted that "natural selection is continually trying to economize every part of the organization" and proposed that instincts, equivalent in modern terms to "genetically programmed neural circuits," arise by natural selection (Darwin, 1859). So our hypothesis breaks no new conceptual ground.

A famous critique of this hypothesis argues that useless features might survive pruning if they were simply unavoidable accompaniments to important features (Gould & Lewontin, 1979). This possibility is undeniable, but if examples are found for neural designs, we expect them to be rare because each failure to prune what is useless would render the brain less efficient—more like Deep Blue—whereas the brain's efficiency exceeds Deep Blue's by at least 10^5.

So what do we claim *is* new? The energy and space constraints have been known for a while, as have various principles, such as "minimize wire." The present contribution seems to lie in our gathering various rules as a concise list and in systematically exemplifying them across spatial and functional scales. When a given rule was found to apply broadly with constant explanatory power, we called it a "principle." Ten are listed as a round number. As with the Biblical Commandments and the U.S. Bill of Rights, some readers

will find too many (redundancy) and others too few. We are satisfied to simply set them out for consideration.

Some readers may object to the expression "design" because it might imply a design*er*, which might suggest creationism. But "design" can mean "*the arrangement of elements or details*," also "*a scheme that governs functioning*." These are the meanings we intend. And, of course, there *is* a designer—as noted, it is the process that biologists understand as natural selection.[3]

Limits to this effort

Our account rests on facts that are presently agreed upon. Where some point is controversial, we will so note, but we will not resort to imagined mechanisms. Our goal is not to explain how the brain *might* work, but rather to make sense of what is already known. Naturally what is "agreed upon" will shift with new data, so the story will evolve. We gladly acknowledge that this account is neither complete nor timeless.

We omit *so* much—many senses, many brain regions, many processes—and this will disappoint readers who study them. We concentrate on vision partly because it has dominated neuroscience during its log growth phase, so that is where knowledge goes deepest at all scales. Also we have personally concentrated on vision, so that is where our own knowledge is deepest. Finally, to apply principles across the full range of scales, but keep the book small, has required rigorous selection. We certainly hope that workers in other fields will find the principles useful. If some prove less than universal and need revision, well, that's science. The best we can do with Data Mountain really is just to set a few pitons up the south face.

1 What Engineers Know about Design

During the Cold War, the Soviets would occasionally capture a U.S. military aircraft invading their airspace, and with comparable frequency a defecting Soviet pilot would set down a MiG aircraft in Japan or Western Europe. These planes would be instantly swarmed by engineers—like ants to a drop of honey—with one clear goal: to "reverse engineer" the craft. This is the process of discovering how a device works by disassembling and analyzing in detail its structure and function. Reverse engineering allowed Soviet engineers to rather quickly reproduce a near perfect copy of the U.S. B-29 bomber, which they renamed the Tu-4. Reverse engineering still flourishes in military settings and increasingly in civilian industries—for example, in chip and software development where rival companies compete on the basis of innovation and design.

The task in reverse engineering is accelerated immensely by prior knowledge. Soviet engineers knew the B-29's purpose—to fly. Moreover, they knew its performance specifications: carry 10 tons of explosive at 357 mph at an altitude of 36,000 feet with a range at half-load of 3,250 miles. They also knew how various parts function: wings, rudder, engines, control devices, and so forth. So to grasp how the bomber must work was straightforward. Once the "how" of a design is captured, a deeper goal can be approached: what a reverse engineer really seeks is to understand the *why* of a design—*why* has each feature been given its particular form? And *why* are their relationships just so? This is the step that reveals principles; it is the moment of "aha!"—the thrilling reward for the long, dull period of gathering facts.

Neuroscience has fundamentally the same goal: to reverse engineer the brain (O'Connor, Huber, & Svoboda, 2009). What other reason could there be to invest 1 million person-years (so far) in describing so finely the brain's structure, chemistry, and function? But neuroscience has been somewhat handicapped by the lack of a framework for all this data. To some degree we

resemble the isolated tribe in New Guinea that in the 1940s encountered a crashed airplane and studied it without comprehending its primary function. Nevertheless, we can learn from the engineers: we should try to state the brain's primary goal and basic performance specifications. We should try to intuit a role for each part. By placing the data in some framework, we can begin to evaluate how well our device works and begin to consider the why of its design. We will make this attempt, even though it will be incomplete, and sometimes wrong.

Designing de novo

Engineers knows that they cannot create a general design for a general device—because there is no general material to embody it.[1,2] Engineers must proceed *from* the particular *to* the particular. So they start with a list of questions: Precisely what is this machine supposed to accomplish? How fast must it operate and over what dynamic range? How large can it be and how heavy? How much power can it use? What error rates can be tolerated, and which type of error is most worrisome—a false alarm or a failure to respond? The answers to these questions are design specifications.

Danger lurks in every vague expression: "very fast," "pretty small," "power-efficient," "error free." Generalities raise instant concern because one person's "very" is another's "barely." To a biologist, "brief" is a millisecond (10^{-3} s), but to an electronic engineer, "brief" is a nanosecond (10^{-9} s), and the difference is a millionfold. Engineers knows that no device can be truly instantaneous or error free—so they know to ask how high should we set the clock rate, how low should we hold the error rate, and at what costs?

The engineer realizes that every device operates in an environment and that this profoundly affects the design. A car for urban roads can be low slung with slender springs, two-wheel drive, and a transmission geared for highway speeds. But a pickup for rough rural roads needs a higher undercarriage, stouter springs, four-wheel drive, and a transmission geared for power at low speeds. The decision regarding which use is more likely (urban or rural) suffuses the whole design. Moreover an engineer always wants to quantify the particular environment to estimate the frequencies of key features and hazards.

One assumes, for example, that before building a million pickups, someone at Nissan bothered to measure the size distribution of rocks and potholes on rural roads. Then they could calculate what height of undercarriage would clear 99.99% of these obstructions and build to that standard.

Knowing the frequencies of various parameters allows rational consideration of safety factor and robustness: how much extra clearance should be allowed for the rare giant boulder; how much thicker should the springs be for the rare overload? Such considerations immediately raise the issue of expense—for a sturdier machine can always be built, but it will cost more and could be less competitive. So design and cost are inseparable.

Of course, environments change. Roads improve—and then deteriorate—so vehicle designs must take this into account. One strategy is to design a vehicle that is cheap and disposable and then bring out new models frequently. This allows adaptations to environmental changes to appear in the next model. Another strategy is to design a more expensive vehicle and invest it with intrinsically greater adaptive capacity—for example, adjustable suspension. Both designs would operate under the same basic principles; the main difference would lie in their strategies for adaptation to changes in demand. In biology the first strategy favors small animals with short lives; the second strategy, by conserving time and effort already invested, favors larger animals with longer lives. As we will see, these complementary strategies account for many differences between the brains of tiny worms, flies, and humans.

Design evolves in the context of *competition*. Most designs are not de novo but rather are based upon an already existing device. The new version tries to surpass the competition: lighter, faster, cheaper, more reliable—but each advance is generally modest. To totally scrap an older model and start fresh would cost too much, take too long, and so on. However, suppose a small part could be modified slightly to improve one factor—or simply make the model prettier? The advance might pay for itself because the device would compete better with others of the same class. A backpacker need not outrun the bear—just a companion—and the same is true for improvements in design. The revolutionary Model T Ford was not the best car ever built, but it was terrific for its time: cheaper and more reliable than its competitors.

How engineers design

An engineer takes account of the laws of physics, such as mechanics and thermodynamics. For example, a turbine works most efficiently when the pressure drop is greatest, so this is where to place the dam or hydro-tunnel. Similarly, power generation from steam is most efficient at high temperatures, which requires high pressures. But using pressure to do work is most efficient when the pressure change is infinitesimally small—which takes

infinitely long. There is no "right" answer here, but the laws of physics govern the practicality of power generation and power consumption—and thus affect many industrial designs.

Similarly, a designer is aware of unalterable physical properties. Certain particles move rapidly: photons in a vacuum (3×10^8 m in a second). In contrast, other particles move slowly: an amino acid diffusing in water (~1 μm in a millisecond)—a difference of 10^{14}. So for a communications engineer to choose photons to send a message would seem like a "no brainer"—except that actual brains rely extensively on diffusion! This point will be developed in chapters 5 and 6.

Designers pay particular attention to the interfaces where energy is transferred from one medium to another. For example, an automobile designed for a V-8 engine needs wide tires to powerfully grip the road. This is the final interface, tire-to-road, through which the engine's power is delivered; so to use narrow, lightly treaded tires would be worse than pointless—it would be lethal. More generally it is efficient to match components—for their operating capacities, robustness, reliability, and so on. Efficient designs will match the capacities of all parts so that none are too large or too small.

Matching may be achieved straightforwardly where the properties of the input are predictable, such as a power transformer driven by the line voltage, or a transistor switch in a digital circuit. But the engineer knows that the real world is more variable and allows for this in the design—by providing greater tolerances, or adjusting the matches with feedback. And to estimate what tolerances or what sorts of feedback are needed, the engineer—once again—must analyze the statistics of the environment. Chapters 8–12 will do this for vision.

What components?

Having identified a specific task, its context and constraints, a designer starts to sketch a device. The process draws on deep knowledge of the available components—their intrinsic properties (both advantageous and problematic), their functional relationships, robustness, modifiability, and cost. A mechanical engineer draws from a vast inventory of standard bolts, gears, and bearings and exploits the malleability and versatility of plastics and metal alloys to tailor new parts to particular functions. For example, Henry Ford, in designing his 1908 Model T, solved the mechanical problem of axles cracking on roads built for horses by choosing a tougher, lighter steel alloyed with vanadium.[3] An electrical engineer solves

an electronic problem by drawing on a parts catalog or else drawing on known properties and costs to design a new chip. Consequently, as models advance, the number of parts grows explosively: the Boeing 747 comprises 6 million parts.

In these respects the genome is a parts catalog—a list of DNA sequences that can be transcribed into RNA sequences ("messengers") that in turn can be translated into amino acid sequences—to create proteins that serve signaling in innumerable ways. This extensive genetic parts list is not the end, but rather a start—for there are vast opportunities for further innovation and tailoring (see chapter 5). An existing gene can be duplicated and then modified slightly, just as an engineer would hope, to produce an alternative function. For example, the protein (*opsin*) that is tuned to capture light at middle wavelengths (550 nm) has been duplicated and retuned in evolution by changing only a few amino acids out of several hundred to capture longer wavelengths (570 nm). This seemingly minor difference supports our ability to distinguish red from green.

At the next level, a single DNA sequence can be transcribed to produce shorter sequences of messenger RNA that can be spliced in alternative patterns to produce subtle but critical variants. For example, alternative splicing produces large families of receptor proteins with subtly different binding affinities—which give different time constants. Other variants desensitize at different rates. How these variations are exploited in neural design will be discussed, as will the capacity to further innovate and tailor the actual proteins by binding small ions and covalently adding small chemical groups (posttranslational modification). In short, with 20% of our genome devoted to coding neural signaling molecules, plus the additional variation allowed by duplication, alternative splicing, and posttranslational modification, the brain draws from a large inventory of adaptable parts. The versatility of these components, as explained further in chapter 5, is a major key to the brain's success.

At a still higher level biological design builds on preexisting structures and processes. Where a need arises from an animal's opportunity to exploit an abundant resource, natural selection can fashion a new organ from an old one that served a different purpose. Famously, for example, the panda's "thumb" evolved not from the first digit that humans inherited from earlier primates but from a small bone in the hand of its ancestors that served a different purpose (Gould, 1992). Thus, efficient designs can be reached via natural selection from various directions and various developmental sequences. This was recognized a century ago by a key founder of neuroscience, Santiago Ramón y Cajal (1909):

We certainly acknowledge that developmental conditions contribute to morphological features. Nevertheless, although cellular development may reveal how a particular feature assumes its mature form, it cannot clarify the utilitarian or teleological forces that led developmental mechanisms to incorporate this new anatomical feature. (edited for brevity)

Neatening up

A designer's list of needed functions might also reveal several that could be accomplished with a single, well-placed component. This has been termed "neatening up," which Ford certainly did for the Model T. For example, rather than manufacture separate cylinders and bolt them together (the standard method), he cast the engine as a solid block with holes for the cylinders. Moreover, rather than use a separate belt to drive the magneto (which provides spark to initiate combustion), he built magnets into the engine's flywheel—thereby reducing parts and weight. The sum of his efforts to improve components (vanadium steel) and design (flexible suspension) and neaten up (engine block, magneto) produced a model that was 25% lighter and delivered 25% more horsepower per pound than the competition, such as the Buick Tourabout.

Brain design reflects this process of neatening up. For example, one synapse can simultaneously serve two different pathways: fast and slow; ON and OFF. One neuron can alternately serve two different circuits: one during daylight and another during starlight (chapter 11). But this strategy must not compromise functionality.

Complicate but do not duplicate

Scientists are constantly lashed with the strop from Occam's razor. That is, we are forcefully encouraged to keep our explanatory models and theories *simple*. So the following design principle seems, not merely surprising, but actually counterintuitive: *if one design is simple and another complicated, choose the complicated* (Glegg, 1969; Pahl et al., 2007). Here is the reasoning: when one part is forced to do two jobs, it can do neither well. An example is the two-stroke engine.

The operating cycle of a four-stroke automobile engine involves four sweeps of the piston through the cylinder. One draws in the fuel, the next compresses it, the third delivers power as combustion drives the piston outward, and the fourth sweeps out the exhaust. The two-stroke engine discharges the exhaust with the same stroke that draws in fuel at the bottom

of the combustion stroke and the beginning of the compression stroke. This serves well for a lawn mower or a power saw because the simpler design avoids the need for a valve gear to separately port fuel and exhaust, giving a better ratio of power to weight. However, the four-stroke engine's more complicated design delivers much more power per liter of fuel and runs smoother and quieter. Moreover, its more efficient combustion discharges fewer pollutants (French, 1994).

There are general advantages to providing a separate part for each task. First, each part can be independently tuned for speed, sensitivity, and so forth—without compromise. Second, each part can be regulated independently. Third, more parts provide more opportunities for further refinement, innovation, and improvement—simply because there are more starting points.

Complicate! is such an important principle for neural design that it seems justified to give one example. The vertebrate retina might have used only one type of photoreceptor but instead it uses two: rod and cone. The rod photopigment is more stable but slower to regenerate, so it serves best in dim light. The cone photopigment is less stable but faster to regenerate, so it serves best in bright light. Having complicated the retina's cellular architecture with two cell types, each type has developed its own molecular refinements—specialized versions of the transduction molecules and of *their* regulatory molecules—all tuned for different light intensities. To take full advantage of these refinements, the two cell types have developed different circuits within the retina. However, just before the retinal output, there is a neatening up: rod and cone circuits merge to share a set of excitatory synapses onto a set of common output cells (ganglion cells). Further explanation is to be found in chapters 8 and 11.

There is another way to complicate a design: include several parts that appear to serve the same function. For example, a neuron may express several enzymes that produce the same product. And neighboring cells may express different versions of a similar protein; for example, axons and astrocytes (glial cells) in optic nerve both express a sodium/potassium pump but with subtly different properties. Also one region may connect to another via multiple pathways: dorsal spinocerebellar tract, ventral spinocerebellar tract, spino–reticulo–cerebellar tract, spino–olivo–cerebellar tract, and so on. These parallel features might once have been regarded as "redundant"—to increase reliability and protect against failure. But now most biologists appreciate that multiple pathways generally serve different roles and thus are not truly redundant.

Indeed, engineers try avoid redundancy, and for good reason. A part waiting for a function occupies space, adds weight, and costs extra. So, this consideration raises suspicion that the multiplicity of intracellular phosphatases, sodium/potassium pumps, spinocerebellar tracts, and so on represent complexity of the good kind.

Choosing materials

Engineers can choose from diverse materials. However, they must try to select the least costly material that is appropriate for the task. For a sailboat mast, wood was traditional, but it is heavy. Graphite can be equally stiff for less weight, but it is brittle; titanium gives the best physical performance, but it is costly. So the choice depends on whether the boat is a dinghy for weekend sailing or a 12 meter yacht for the America's Cup.

Brain design is forced to select from a far narrower set of materials. For example, biological membranes are composed of lipids and proteins. Although mechanisms for regulating the passage of substances and ions *across* the membrane in either direction are myriad (ion channels, pumps, cotransporters, antiporters, flippases, etc.), the intrinsic properties of the membrane itself are relatively constant. In particular, the membrane's specific capacitance is fixed at around 1 $\mu F\ cm^{-2}$. Neurons generate electrical signals by opening and closing channels in the membrane that allow ions to move down their electrochemical gradient and carry charge in and out of the cell.

The time constant of this electrical response is the product of the membrane resistance and capacitance, but capacitance is fixed. Therefore, to speed up an electrical process, a neuron of given surface area must reduce its membrane resistance by opening more channels, thus allowing more ions to cross the membrane. The cost of restoring these ions so as to maintain the electrochemical gradient is high—in fact, it is the human brain's major energy cost: more than 60% goes for pumping ions, making this a key constraint upon design. Thus, for the brain, as for 12 meter yachts and automobiles, speed comes at a premium—and the brain is forced to use it sparingly. This theme will recur.

Integrating across systems

Engineers look for trade-offs among individual components to improve overall performance. For example, because a truck's suspension reduces shock, investment in better springs and shock absorbers can be traded for

weight and strength in the axles. So designers evaluate the whole system to discover where investment in one component is more than compensated by savings in others. This integrated approach to design extends the principle of matching components to include cost.

An example relevant to neural design is the mobile telephone (Mackenzie, 2005). Like many animals, it is small and roams on limited power. Models compete fiercely, and success depends upon performance, beauty, and energy efficiency. One notable innovation provides the phone's "brain," its tiny internal computer, with a *turbo code* that extracts wireless signals from environmental noise. This code employs an algorithm for *belief propagation* that is computationally expensive. But the investment pays because the code eliminates noise so effectively that the efficiency of wireless communication approaches the theoretical limit defined by Shannon's equation (chapter 5).

Optimizing efficiency allows the phone to reduce the amplitude of its output signals. These consume the highest proportion of the phone's power because more energy is needed to transmit radio signals long distances in all directions than to send electrical pulses along short connections in a tiny computer. Consequently, the energy invested in the phone's brain for turbo coding produces much larger savings in the heavy work of signal transmission. By analogy, an animal's small brain saves energy by efficiently directing the activities of large, power-hungry muscles.

To understand the design of an integrated system requires teamwork. When no single person can grasp the details of every component and process, designers team up. Specialists integrate their detailed knowledge of each particular into an efficient whole. It was a team of specialists that reverse engineered the B-29. They needed to combine expertise in aerodynamics, structural engineering, materials science, fluid mechanics, control systems, and so on. Neuroscientists are reaching the same conclusion and forming teams that integrate specialized knowledge to reverse engineer their systems. Brains are integrated systems because they evolved to integrate, so how else can we understand them?

How to proceed and a caution

To consider brain design as a problem of reverse engineering, we must begin with an overview of its main tasks, establish some basic measures of performance, and then see how these relate to the investment of resources in particular mechanisms (chapters 2 and 3). Having established some basic principles, we select one important system—vision—and treat each stage of

processing in the framework of design. We present the environmental context, then the circuit structure and some "hows" of its functioning. Then for each stage we will point out some "whys" of the design and note where other neural systems use similar principles.

The design principles evinced here do not explain everything. In fact, principles cannot explain how anything works—not the B-29, not the Model T, and certainly not the brain. What, then, is their use? Design principles deepen our understanding of why things work the way they do, and armed with this deeper understanding, we can reverse engineer more efficiently. Of course, applying inappropriate or misguided principles would slow us down. Thus, principles derived theoretically, without real objects and mechanisms to illustrate them, are not yet of much use. So we attempt to balance the insights that come from principled explanations against the doubts that come from overdoing them.

2 Why an Animal Needs a Brain

Essentially only one thing in life interests us: our psychical constitution. The considerations which I have placed before you employ a scientific method in the study of these highest manifestations in the dog, man's best friend.

—Ivan Pavlov, Nobel lecture, 1904 (edited for brevity)

Brain books generally begin at the lowest levels—neurons, axons, synapses, and ion channels. But that approach ill suits our goal of reverse engineering. One cannot explain a B-29 by starting with the nuts and bolts. So we postpone the parts lists and detailed schematics to consider first a larger question: why do we *need* a brain?

One's first thought, of course, is that we need it for the magical activities and feelings it confers: art, music, love . . . consciousness. But although these features arouse intense curiosity—as Pavlov emphasized—we shall see that they are merely baroque decorations on the brain's fundamental purpose and should not be mistaken for the purpose itself. What we identify here as the brain's purpose, especially because we are seeking principles, should apply not only to humans but as well to the nematode worm, *C. elegans*, and to flies. The deep purpose of the nematode's brain of 302 neurons, the fruit fly's brain of 10^5 neurons, and our own brain of 10^{11} neurons (Azevedo et al., 2009) must be the same. By identifying the basic purpose, we set a context for later considering the "decorations." We expect that research on the mammalian cerebral cortex will not reveal many new principles—rather it will elaborate the core ones. In general, it should be easier to discover them in simpler brains.

The brain's purposes reduce to regulating the internal milieu and helping the organism to survive and reproduce. All complex behavior and mental experience—work and play, music and art, politics and prayer—are but strategies to accomplish these functions. Sharing these fundamental tasks, the brains of worms, flies, and vertebrates show significant

similarities—which will be discussed. But first, consider that a tiny bacterium, *E. coli*, and a much larger single-celled protozoan, *Paramecium,* manage these two tasks quite well without a brain. How?

Lives of the Brainless

A bacterium foraging

E. coli is miniscule (1 × 3 μm) and thrives in a nutritive soup—adrift in the intestinal digests of a large animal (figure 2.1; Alberts et al., 2008). The microbe is equipped with "taste" receptors, a battery of proteins each of which specifically binds an attractant (such as an amino acid or sugar) or a repellant. These receptor proteins cluster on the surface membrane and form signaling complexes within which they cooperate to increase sensitivity and response speed. The largest cluster is at the forward end ready to taste what comes as the bacterium ploughs through the soup. Although each cluster comprises thousands of molecules—to increase the chance of catching a taste—there are only five types of receptor molecule, each responding to a range of related compounds.

The first function of these receptors is to evaluate the *soup du jour*. Each potential nutrient (amino acid, sugar, etc.) requires its own specific transporter (*permease*) for uptake into the bacterium, plus a particular enzyme or even a whole set of enzymes to process it for energy and materials for growth. It would be uneconomical to maintain high levels of all possible transporters and processing enzymes when only a subset is needed at a given moment. Therefore, a cell refrains from synthesizing proteins for uptake and digestion until a taste receptor binds the target molecule. A receptor's binding affinity determines the concentration at which protein synthesis becomes economical.

For its default fuel *E. coli* uses glucose. But when glucose is off the menu, it can use lactose. This requires lactose detectors to call for two proteins: a permease to admit lactose and an enzyme, galactosidase, to split it. The genes coding these proteins are adjacent in *E. coli*'s DNA, comprising an *operon* (genes that work together). Their expression is blocked by a repressor protein that binds to this stretch of DNA and blocks the entry of RNA polymerase, the molecular machine that transcribes DNA to RNA (*RNA polymerase*) to initiate protein synthesis (figure 2.2). The repressor *is* the lactose detector which, upon binding allolactose (an isomer that always accompanies lactose) changes shape and releases from the DNA. This allows RNA polymerase to move off and transcribe the operon (figure 2.2; Phillips et al., 2009).

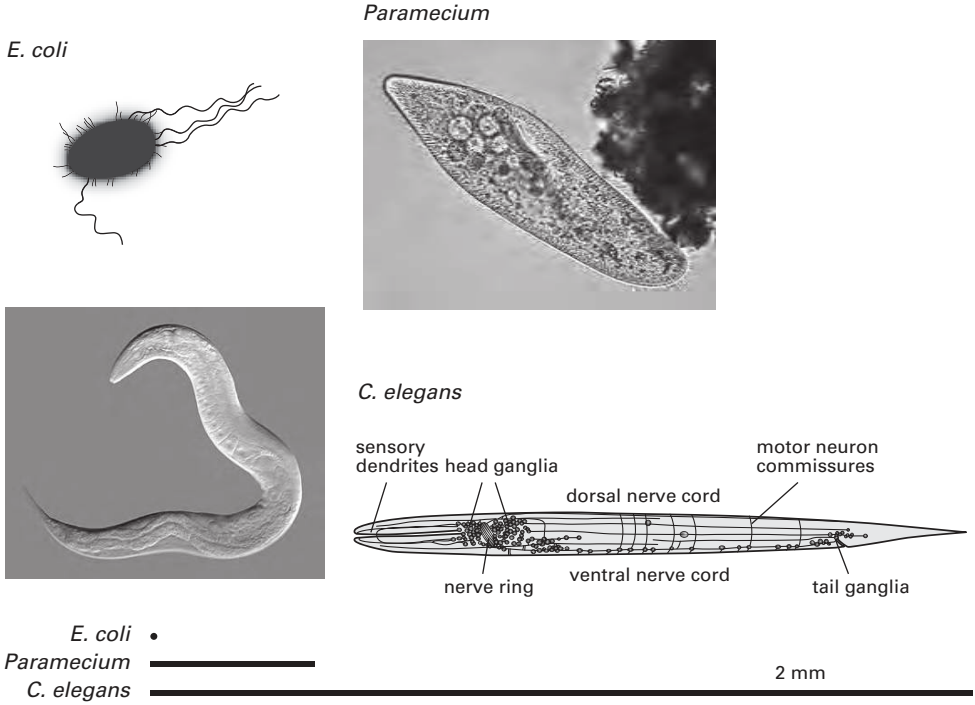


Figure 2.1

Three organisms of increasing size: bacterium, protozoan, and a nematode worm. Note the different scales: micrometers to millimeters. Body lengths are drawn to the same scale at the bottom of the diagram. *Paramecium caudatum* and *C. elegans* photos are light micrographs of live specimens. Diagram of worm indicates the positions of neurons that form the brain. Light micrographs from Wiki commons. *C. elegans* from Wikimedia Commons, CC BY-SA 3.0 / Bob Goldstein, UNC Chapel Hill, http://bio.unc.edu/people/faculty/goldstein/. *Paramecium* by Alfred Kahl, public domain, from Wikimedia Commons.

In effect, the lactose receptor *predicts* for the organism what it will need to exploit this new resource. By encoding the permease and the digestive enzyme together, one sensory signal can evoke all necessary components in the correct ratios. Thus, a given level of lactose in the soup calls for the proper amount of permease which is matched by the proper amount of galactosidase. This design principle—matching capacities within a coupled system—is a key to the organization of multicellular animals where it is called "symmorphosis" (Weibel, 2000). We see here that symmorphosis begins in the single cell.

The lac operon

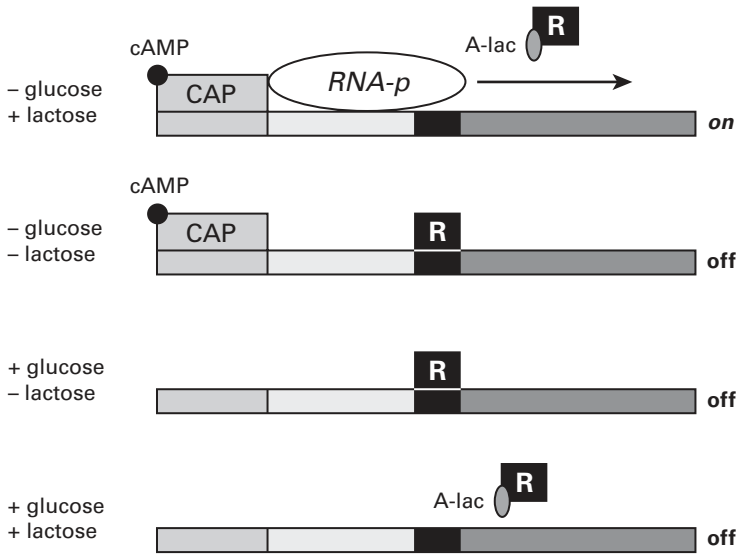


Figure 2.2

The lac operon: a molecular mechanism that discriminates between patterns of input and determines action. To transcribe the lac operon's genes, RNA polymerase (*RNA-P*) must bind to its site and move into the operon's DNA. Its movement is blocked by the repressor R, but R cannot bind and block when holding a molecule of allolactose (A-lac). To start moving, *RNA-p* must be activated by the protein CAP. This activator protein only binds to its site on the DNA when it is binding cAMP, and cAMP is eliminated in the presence of glucose. Thus, *RNA-p* only transcribes the lac operon when glucose is absent and lactose is present.

On occasions, such as when its host has eaten an ice cream, *E. coli* is presented with both lactose *and* glucose. Now the bacterium need not metabolize lactose and so need not build machinery to process it. To block this futile activity, there is a second molecular switch. RNA polymerase, to step along the DNA transcribing the lac operon, must be activated by the protein CAP, and CAP must be binding a small signaling molecule, cAMP. Biochemical pathways couple the production of cAMP to the concentration of glucose. As glucose rises, cAMP falls; this turns off the RNA polymerase (figure 2.2), and *E. coli* stops producing unneeded machinery.

Thus, a molecular control system combines information from two inputs to compute the correct conditions for processing lactose: IF lactose AND NO glucose, then GO; IF lactose AND glucose, then NO GO. The chemical

network controlling the lac operon enables a single cell to detect specific patterns of events and to mount concerted patterns of response that promote survival and reproduction. Of course, this is what a brain does on a larger scale, and in doing so it builds upon the capacities for executing logic that reside in the molecular control systems of single cells (Bray, 2009).

E. coli does more than just taste the soup and reprogram its digestive enzymes. The taste receptors also direct the cell to forage, that is, to discover and migrate to regions of higher nutrient concentration. To execute this process, *chemotaxis*, the bacterium propels itself with flagella, which are helical screws that rotate at 6,000 rpm. Their beating sends it tumbling off in random directions for brief periods, each followed by a short, straightish run. A surface receptor, sensing the instantaneous concentration of a nutrient, compares it to the past concentration—"past" lasting 1 s. If the new concentration is higher, the motor apparatus holds the forward course for a bit longer.

This search strategy (*biased random walk*, figure 2.3) resembles the party game where an object is hidden and a searcher is simply told "warmer . . . cooler . . . warmer, warmer. . ." The mechanism can sum signals from several attractants—maintaining the direction of motion for a longer time. Or, it can sum antagonistic signals (attractant + repellent) and change direction sooner. Thus, with a sensor, plus a "working memory" that controls a propeller, a microbe's wandering eventually delivers it to a greener pasture (Berg, 1993).

A microbe's memory

E. coli's working memory is simple: it is imprinted on the receptor protein by means of a negative feedback loop. The activated taste receptor causes an enzyme to attach methyl groups to the receptor complex, decreasing its sensitivity. The number of methyl groups on a receptor indicates how strongly it has been activated, and because the feedback loop is sluggish, the record stretches back into the bacterium's frantic past—1 s. The mechanism, by using the past to set receptor sensitivity, determines the bacterium's response in the present—a reasonable definition of memory. Thus, a single cell can store information cheaply through chemistry—by covalently modifying a signaling molecule.

In accomplishing the basics (preserve internal milieu and reproduce), this single cell uses mechanisms that are either optimal or highly economical: just the right number and distribution of taste receptors, just the right ratios of transporters and digestive enzymes, just the right levels of protein expression to match costs versus resources, plus the smallest signaling

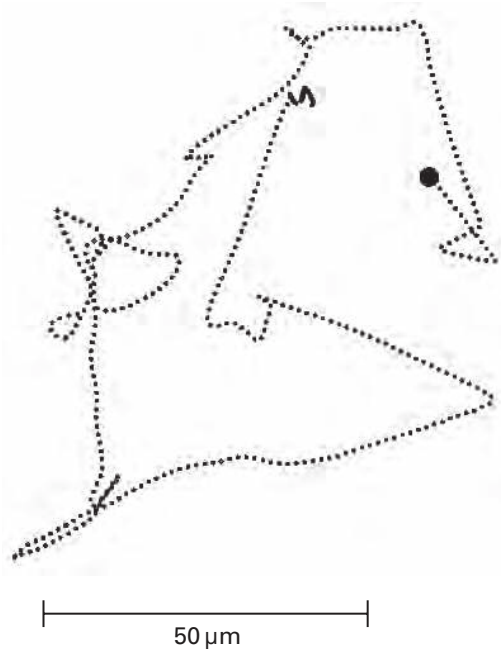


Figure 2.3
***E. coli*'s biased random walk**. By moving forward more and turning less, as the concentration of attractant increases, *E. coli* approaches the attractant's source. Tracing shows 26 runs over about 30 s with a mean speed of 21.2 µm/s. Reprinted with permission from Berg and Brown (1972). For videos of *E. coli* swimming see http://www.rowland.harvard.edu/labs/bacteria/index_movies.html/.

network for chemotaxis that could provide sufficiently robust performance. Moreover, its working memory suffices to steer the motor toward food and mates. Although a memory lasting only 1 s may not seem impressive, realize that to store a long history of lactose concentrations would be pointless—because they are themselves evanescent. Given its lifestyle, the bacterium's memory is just about as long as it *should* be.

This microbe easily lives like a Zen master—in the moment. Feed the cell, and in an hour it is gone, divided among its progeny. But once an organism becomes large enough for a brain, the Zen injunction—"Live in the moment"—itself becomes a Zen koan. A brain provides the organism with a more significant individual past and a more extended future with which to exploit it. But so equipped, staying in the moment becomes as unimaginable as the sound of one hand clapping.

Limitations to life as a microbe

Given that bacteria accomplish the basics so well, one must consider the limitations. First, their ability to respond to environmental challenge resides largely in genetic memory. A *population* thrives by reproducing rapidly and exchanging genetic material—so that when the environment changes, at least one individual in the population will contain a gene to deal with it. Thus, a population can "learn" to exploit new resources—such as potentially delicious industrial waste. However, an individual microbe, suddenly losing glucose in a lactose-rich medium, can respond only if its genome already contains the lac operon.

Second, an individual microbe cannot actively move very far. It can neither return to the site of its last meal nor deliberately transfer to a new host. This confines each species of microbe to the restricted environment for which it has specialized: a termite's gut or the skin of a human inner elbow (Grice et al., 2009)—where the bacterial genome is prepared for what it will likely encounter, and where surprises are relatively few. But this leaves a wider world unexplored and thus unexploited.

To explore would certainly increase the chances of encountering a more favorable medium—but there is a limiting challenge: size. For such a miniscule object, water is tremendously viscous. Top speed for *E. coli* is 30 μm per second, and when its effort ceases, there is insufficient inertia to carry it forward, so it abruptly stops within 0.01 nm (chapter 5; Purcell, 1977; Nelson, 2008). For a human it would be like swimming in thick molasses—agonizingly slow and energetically expensive. Consequently, to move over long distances, bacteria have evolved other methods, for example, by being sticky and hitching rides on animals.

In short, a bacterium inhabits a tiny universe—barely a few centimeters—where the critical factors are beyond its control. When transportation relies on random, energetically expensive self-propulsion or the kindness of strangers, life is precarious. A cell that could propel itself more rapidly and cheaply could forage more widely, but to overcome the effects of Brownian buffeting and high viscosity it must enlarge. And it need not get very large before motor coordination becomes an issue—as we now explain.

Protozoa: bigger and faster but still brainless

Paramecium, the familiar single-celled protozoan, measures up to 350 μm × 50 μm. Being 300,000-fold larger than *E. coli*, it is less subject to viscous forces. *Paramecium* propels itself with cilia that cover its surface and coordinate their beating to send synchronous waves from head to tail. Cruising

speed can reach roughly 1,400 μm per second, 50-fold faster than *E. coli* and with lower relative energy cost. In human terms this is the difference between exploring on foot at 4 mph and racing a car at 200 mph. Consequently, *Paramecium* can explore relatively enormous volumes of pond water and harvest bacteria by sweeping them into its "mouth." This microshark is guided by a variety of taste receptors to approach sites where bacteria proliferate, for example, clumps of rotting vegetation. It also has nociceptors to detect toxic sites, such as overripe sludge contaminated with hydrogen sulfide.

In its cluttered environment *Paramecium* inevitably encounters immovable obstacles, and to avoid the futility of continual ramming, *Paramecium* has evolved a useful response (figure 2.4; Jennings, 1904; Eckert, 1972). At the first bump it throws its cilia into reverse and backs off by a few millimeters. Then it does a quick twiddle, switches to forward, and sets off in a new direction. This avoidance response is fast—completed within a fraction of a second—and it has to be. Futile activity wastes time and energy; moreover, the immovable object might be a predator!

E. coli's chemical signaling systems could not trigger and coordinate this rapid response. Diffusion suffices for *E. coli* because the distance is short—a small intracellular messenger molecule diffuses throughout the bacterium in about 4 ms. But diffusion time increases as the distance squared (Nelson, 2008), so for a *Paramecium* that is 100-fold longer than *E. coli*, diffusion from "head" to "tail" would be 10,000-fold slower, about 40 s. Obviously, this is far too slow for receptors at the head to call "*Reverse!*" to the tail cilia. Electrical signals spread much faster: a change in membrane voltage initiated at the head reaches the tail in milliseconds.

Electrical signaling for this avoidance response requires several new components. First, a mechanoreceptor is needed to detect the bump. This involves a specialized cation channel inserted into the cell membrane. Stretch on the membrane deforms the channel, opening it to sodium ions that rapidly depolarize the membrane (<100 μs). Depolarization opens voltage-sensitive calcium channels that admit a rush of calcium ions—further depolarizing the membrane, opening still more calcium channels, and so on. This positive feedback produces a robust response that recruits calcium channels across the entire membrane (figure 2.4). They open briefly, then close and inactivate. Thus, the two components—stretch-gated sodium channel plus voltage-gated calcium channel—cooperate to deliver a synchronous pulse of calcium over the cell's entire surface.

The reason to spread the electrical signal via a calcium channel, rather than a voltage-gated sodium channel (such as used by nerve and muscle), is

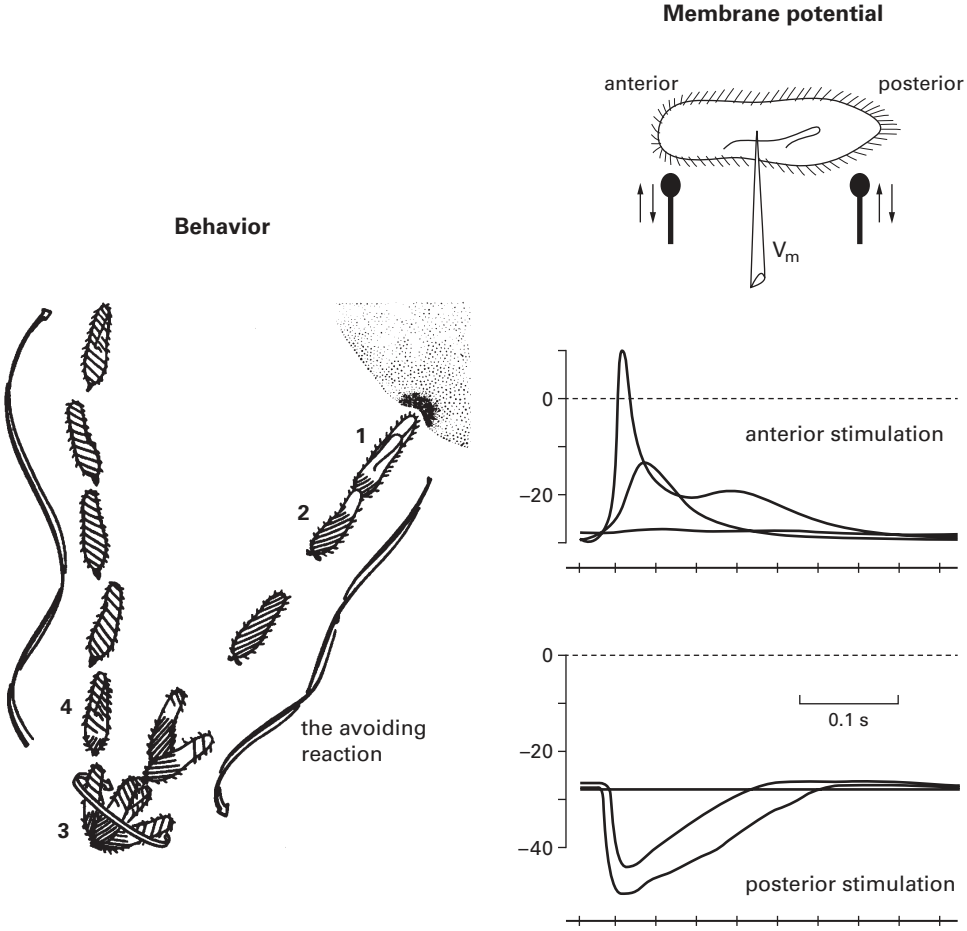


Figure 2.4

***Paramecium*'s avoidance response: behavior and electrical mechanism**. **Left**: The four stages of behavior. (1) Bumps up against immovable object, (2) backs off by reversing cilia, (3) gyrates while cilia switch from reverse to forward, and (4) sets off in a new direction. **Upper right**: Measuring electrical response to mechanical stimuli. Intracellular microelectrode records membrane potential and probes prod the membrane. **Middle right**: Membrane potential recorded following stimulation with anterior probe. A weak prod depolarizes membrane for 300 ms (lower trace). A strong prod generates a short calcium action potential followed by longer depolarization (upper trace). **Lower right**: Posterior prod hyperpolarizes. The response to the weaker prod is smaller and has a longer latency. Adapted from Eckert (1972), with permission.

that a calcium ion can also serve intracellularly as a chemical messenger. In this case the chemical message arrives synchronously at the base of all cilia, saying "*Reverse beat*," and their simultaneity adds power to the reversal. As *Paramecium* backs up, calcium pumps in the membrane vigorously reduce the calcium level, allowing patches of cilia to slip back into "forward"—explaining the indecisive twiddle. Once most of the calcium has been extruded and all cilia again beat forward, *Paramecium* heads off in a new direction (figure 2.4).

The system is polarized. The stretch channels are at the head, ensuring that the calcium pulse that reverses the cilia will also reverse the animal. The decision to reverse is structured as a simple threshold: when a bump is sharp enough, stretch channels open sufficiently to depolarize the membrane smartly enough to kick the calcium channels into their regenerative cycle. The numbers and sensitivities of stretch channels are adjusted to discriminate a truly immoveable obstacle from a yielding one. Conceivably, they are even tuned by experience via the attachment of some chemical group as with *E. coli*'s working memory.

Finally, the twiddle that sets *Paramecium* off in a new direction occurs because some patches of cilia enter forward gear before others, perhaps by the molecular noise in calcium pumps (chapter 6). Whatever the exact mechanism, the twiddle generates a random direction—which is good. Lacking distance receptors, *Paramecium* cannot predict which search direction is most likely to be best, so random behavior is optimal (Reynolds & Rhodes, 2009). Also, random motion prevents a predator from predicting *Paramecium*'s next move, thus making it harder to catch.

Where brains emerge

Despite the advantage of its fast control system for locomotion, *Paramecium*'s behavioral repertoire is limited. One impediment to richer behavior is that there is only one cell membrane and thus only one line for fast (electrical) communication. But more deeply, the cell is still so small that locomotion must be slow, and the environment remains so evanescent that richer behavior and longer memory offer no advantage. *Paramecium*'s exploitable world remains sufficiently restricted that one communication channel is plenty. Multicellularity can pay—but only when an animal becomes slightly larger and lives slightly longer in an environment where clues to food and danger persist.

The crossover—where multicellular animals arise and dominate (eat the unicellular)—occurs at a size of around 1 mm and a lifetime of days.[1] Then

cells specialize and associate to form tissues, tissues form systems, and systems cooperate to form a more versatile organism. Thus, multicellularity follows the engineering principle *complicate* (Glegg, 1969/2009a). The many tasks performed by a single cell are now divided among many specialized components. Naturally, coordination is required at each level (cell, tissue, organ, system, and organism) and across levels.

Coordination demands some mechanism with an overview that enables it to weigh alternatives, set priorities, and then exert ultimate authority to execute. Fortunately, the multicellular design that demands such integration also provides a special class of cells to accomplish it. These cells—neurons—now do what *Paramecium* could not: provide multiple fast lines for communication. In short, for a multicellular organism a brain becomes necessary, possible, and profitable.

Worm with tiny brain

The nematode worm, *C. elegans*, measures about 1×0.1 mm (figure 2.1) and in its predominant hermaphroditic form comprises exactly 959 somatic cells (Herman, 2006). It lives close to the soil surface and feeds on bacteria in rotting vegetable matter. Unlike *Paramecium*'s pond water chemicals in soil and humus are not swept away by convective currents—they move by diffusion and capillarity through a matrix, so traces persist (Félix & Braendle, 2010). The matrix and surface film provide firmer substrates for locomotion, and these allow the worm's sinuous crawl to open up whole new continents for exploitation.

The worm's enlarged territory and its locomotion through a labyrinthine matrix with persistent chemical traces warrant an upgrade. The worm improves the chemotaxis system and adds diverse sensors (of current state, opportunity, and danger), plus a larger repertoire of behavioral responses and a longer memory (de Bono & Maricq, 2005). Because bacteria-rich patches are oases where many species compete, the worm's success requires that it move smartly across a patch to efficiently find and exploit the productive regions, meet, mate, and lay eggs.

Improved foraging must be matched by more efficient systems for digestion, absorption, metabolic storage, and elimination. And as the behavioral repertoire expands, there is more need to evaluate and prioritize. For example, upon encountering a good hunting ground, how much heat or acidity should it tolerate? Upon encountering two chemical traces, which should it follow? When to search and when to graze? When to mate and when to be

stilled by "satisfaction"? In short many of the choices posed for humans by Ecclesiastes arise even for this apparently simple worm—which decides with its tiny brain.

The worm's brain may be small, but its 302 neurons plus 56 glial and support cells comprise nearly 40% of its body's entire complement. The figure in humans is close to 1%. So we first consider some behavioral advantages that justify its immense investment. Then we consider the brain's design, noting the features shared with larger brains that suggest they are governed by principles of neural design.

Locomotion

Grazers must keep on the move. The worm moves forward by bending just behind the head and then propagating the bend toward the tail. Driven by this sinusoidal wave, it threads its way through soil and rotting vegetable matter, swims through pools of fluid, and crawls across moist surfaces (e.g., decaying fruits, agar plates in laboratories). A worm travels fastest when rigid objects are regularly spaced at 0.5 mm (figure 2.5), and if this spacing is changed by just 10%, their forward speed halves. A worm seems designed to cope best with the average particle size in its preferred habitat, like a pickup truck designed for rough roads (Park et al., 2008).

But *C. elegans* is both truck and driver, continually adapting its propulsion to cope with changing conditions. When the worm goes from swimming in a pool to crawling across a wet surface, the surface tension increases viscous forces 10,000-fold, and the worm adjusts its undulations accordingly (figure 2.5). Frequency falls tenfold, wavelength shortens threefold, and more muscular power is transferred to the viscous medium. The worm continuously adjusts its drive train over a wide range of conditions, maintaining the wave's angle of attack at an efficient value, close to 45° (figure 2.5). To understand how, we must examine the integrated locomotor system: brain, muscles, body, and substrate.

A sequence of muscular contractions produces the moving wave (Sengupta & Samuel, 2009). Muscle cells on the upper side of the body contract to bow out the lower side, and when the upper cells relax, the body springs back, driven by an internal hydrostatic pressure of 0.5 atmospheres. The wave is propagated by sending two opposite bends along the body, one after the other (figure 2.6), and this sequence repeats at the frequency of undulation. When the head leads the tail, the wave moves down the worm, pushing it forward, and when the tail leads, the worm moves backward. The head also wags from side to side, and when the worm decides to

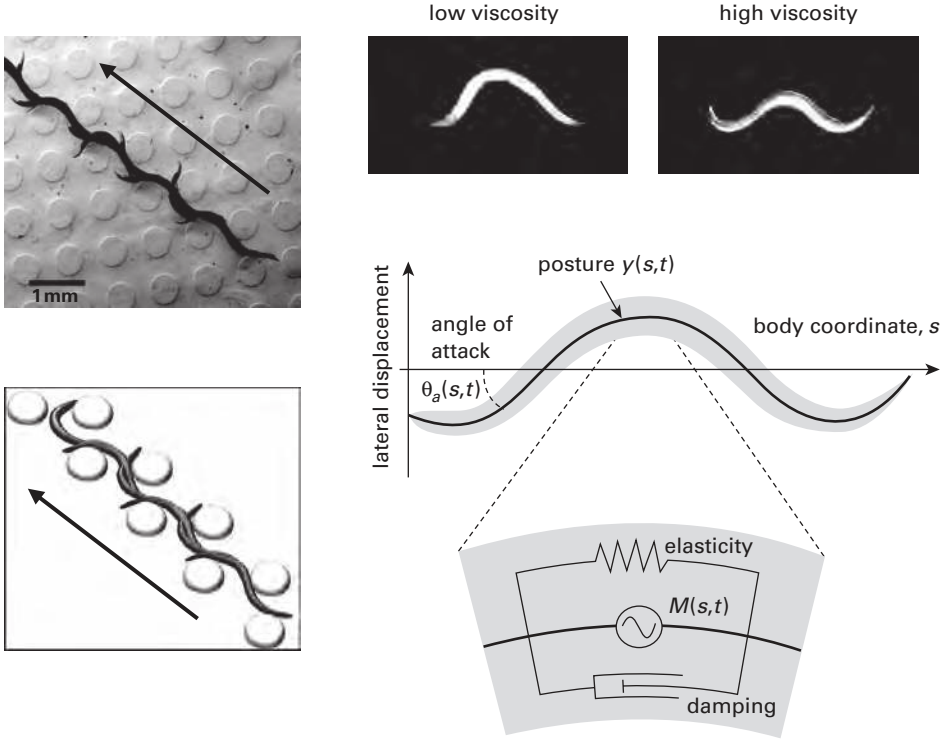


Figure 2.5

***C. elegans* locomotion matches the terrain and adapts to viscosity**. Spacing of soil particles affects forward speed, as shown when worm crawls through a regular array of agar posts of given spacing. **Upper left**: Superposition of 10 photos taken at 200-ms intervals as a worm traversed the array in which it moved forwards at maximum speed. **Lower left**: Tracings of five of the above photos, taken at 400-ms intervals, show why speed is maximum: body wavelength matches post spacing to distribute thrust efficiently. **Upper right**: The wavelength of undulation is longer in a low-viscosity medium and shorter in high viscosity. **Middle right**: Body posture is described by $y(s,t)$, the lateral displacement, y, changing with position along body, s, and time, t. The angle of attack at a given position and time, $\theta_a(s,t)$, is critical for determining thrust against the substrate. **Lower right**: The factors determining body posture and its dependence on viscosity. These vary with position along the body, s, and change with time t. In a simple biomechanical model the muscle force $M(s, t)$ interacts with body elasticity and viscous damping by the medium, to determine lateral displacement $y(s, t)$ and the angle of attack $\theta_a(s, t)$. Left reprinted with permission from Park et al. (2008). Right reprinted with permission from Fang-Yen et al. (2010).

suddenly change direction, it bends the whole body and then springs back—a good tactic for evasion and escape.

These four distinct patterns (forward, reverse, wag, and turn) are produced by 75 motoneurons that control 95 muscle cells. Each muscle cell receives input from one excitatory and one inhibitory neuron which are activated in strict alternation (Bullock, Orkand, & Grinnell, 1977). To bend the head, an excitatory motor neuron on one side of the body activates a muscle, and an inhibitory motor neuron suppresses the corresponding muscle on the other side. To propagate the bend as a wave, motor neurons activate sequentially along the body. Their output frequency determines the frequency of the undulation, and their phase determines its waveform. Excitatory motoneurons on one side activate with inhibitory motoneurons on the opposite side and alternate with excitatory motor neurons on that side (figure 2.6). Where should one look for the oscillators that produce these cycles of motor neuron activity?

Search for the oscillators

Early studies of animal locomotion were fraught with bitter argument about the origins of cyclical activity—such as stepping. Oscillations might be produced within the nervous system by local circuits (*central pattern generators*). Or they might be produced outside the nervous system by cycling sensory feedback (Marder & Bucher, 2001; Goulding, 2009). The feedback mechanism was proposed early for vertebrate stepping. One set of motor neurons excites muscles that extend the limb. This activates sensors that inhibit the extensor motor neurons and excite the flexor motor neurons, thus retracting the limb. Flexion activates sensors that inhibit the flexor motor neurons and excite the extensor neurons, and so on.

Many animals combine the two mechanisms. A central pattern generator sends cyclical commands to the motor neurons, and sensory feedback adjusts their phase, frequency, and amplitude to match changes in external load (Burrows, 1996). But the worm's circuitry seems not to use a central pattern generator. No intrinsically oscillating neurons have been found, nor does the brain's wiring diagram (see below) show the typical oscillatory circuit—a small group of neurons that send signals around a closed loop. Worms are capable of making central pattern generators—some of their cells use internal biochemical oscillators to control the rhythmical movements of ingestion, defecation, and copulation. That the worm can make central pattern generators but does not do so for locomotion suggests that it might have found a better way. Rather than relying on a pattern generator in its brain, the worm exploits its body.

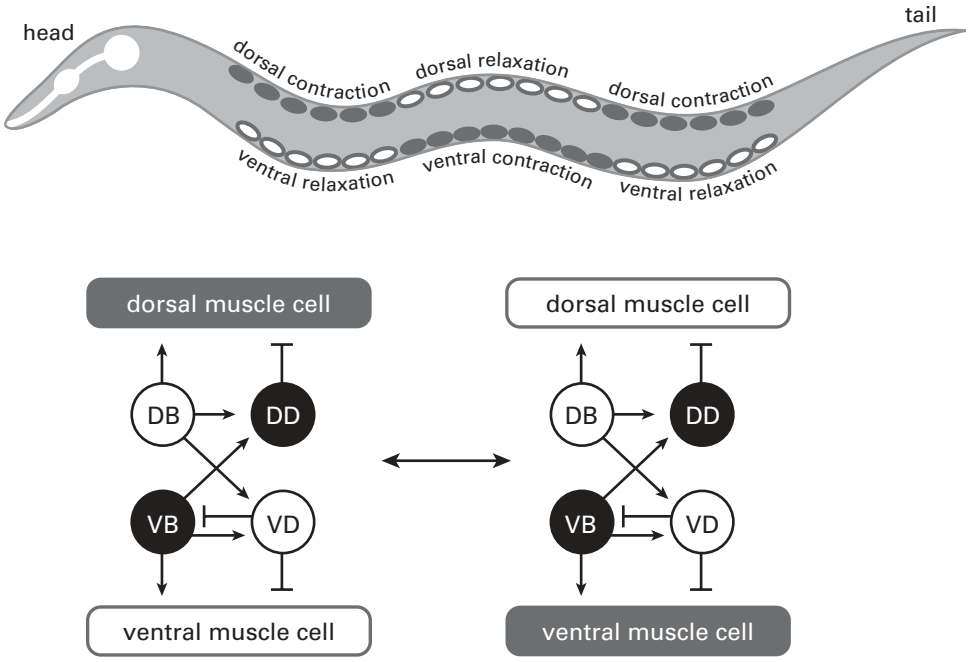


Figure 2.6

Neural circuit that bends the worm. Excitatory motor neurons (DB, VB) alternately cause dorsal and ventral muscles to contract, whereas inhibitory motor neurons (DD, VD) alternately cause them to relax. The excitatory motor neuron on one side drives the inhibitory neuron on the other side so that the body bows downward (DB and VD active), or upward (VB and DD active). This cross-inhibitory circuit repeats along the worm to promote a traveling wave. Modified from Sengupta & Samuel (2009), with permission.

Cycling with the body

The worm builds its oscillator by combining feedback with body mechanics. A burst of activity in motor neurons drives the muscles on one side. Their contraction bends the body and tensions the body's intrinsic spring—internal hydrostatic pressure. Sensors excited by these forces feed back to inhibit motor neurons, whereupon the muscles relax and the body springs back. This terminates the negative feedback, allowing the motor neurons to reactivate and start a new cycle (figure 2.6). Because the spring is damped by viscous forces (figure 2.5), the oscillation is well behaved. Also, it automatically adjusts to changes in viscous load, smoothly shifting the worm's gait to match operating conditions.

So by using its biomechanics the worm can dispense with a central pattern generator, thus freeing up brain space. Here, then, is a useful design principle for motor systems: lighten the brain's load by using the body. Engineers call this *embodied computation* (also embodied intelligence or cognition; Pfeifer & Bongard, 2006).

In the early days of robots, crawling and stepping movements were generated by an all-powerful central computer—an omniscient central pattern generator. This artificial intelligence collected sensory information and fed it into a complicated program that, by modeling the robot's mechanics, worked out the necessary commands and sent them to slavish limbs. To implement this top-down design required the robot to drag around a heavy computer, which, in turn, meant thicker limbs and stronger actuators—the result, a power-hungry behemoth. It was eventually realized that the robot and its limbs *are* a computer, an analogue computer that runs its mechanics in real time (Brooks, 1990). This analogue computer comes for free and can be set up to process information for control by, for example, being part of an oscillator. This insight inspired a new generation of small, efficient, and adroit stepping machines that blew away the behemoths. Thus, the worm exemplifies embodied computation with a neuromechanical system that matches and integrates a few basic components to meet specifications efficiently.

Neural circuits coordinate patterns of movement

Despite the contribution of body mechanics to the oscillator, neural circuits are still essential—they close the loop inside the worm. The neural circuits must be correctly configured and tuned to work with the biomechanics. Sensors must give the right feedback to motor neurons, and motor neurons must send the right signals to the right muscles with the right timing. Circuits are constructed to make this happen by ensuring that as muscles on one side of the body contract, the antagonistic muscles on the other side relax: motor neurons on one side inhibit the excitatory motor neurons for the antagonists and also excite their inhibitory motor neurons (figure 2.6). Here, then, is a circuit motif, *reciprocal inhibition* (Sherrington, 1906), that is widely employed in brains because it simply and effectively solves a common problem.

Changing direction

The brain produces motor rhythms for "forward" and "backward" using two separate sets of motor neurons. Each set has its own circuit: one works with the biomechanics to send the undulatory wave head-to-tail and the

other works to send the wave tail-to-head. This is not a popular design. Most animals use a single set of motor neurons as the final common pathway for all commands to muscle. Using two independent sets, each with a full complement of connections and synapses to muscles, seems wasteful, so why does the worm do this? We speculate that for a small brain with neuromechanical oscillation, two sets of motor neurons are cheaper than a complicated central pattern generator.

Directing action

Like *E. coli* and *Paramecium*, the worm acts to improve its chances of completing its production on the ecological stage. Equipped to move further and faster, its costs are higher and the risks greater, but so are the opportunities and rewards. So the acts must be directed appropriately (de Bono & Maricq, 2005; Lockery, 2011).

The simplest acts are aversive responses, similar in purpose and effect to *Paramecium*'s avoidance response. Tap the worm's head, and it immediately wriggles backward; tap its tail, and it wriggles forward. Two simple circuits generate this behavior (figure 2.7). Mechanosensory neurons at the front drive interneurons that activate the "backward" set of motor neurons, and mechanosensory neurons at the rear drive interneurons that activate the "forward" set. The two sets have cross connections to prevent their working in opposition.

Just as the purpose of *E. coli*'s actions is laid out in chemical circuits in a single cell, so the purpose of the worm's behavior is laid out in the connections between neurons. Naturally a brain with many neurons can generate richer behavior because, by forming connections between cells, it makes more circuits. How has the worm's brain harnessed this potential and moved its behavior beyond the simple reactions of *E. coli* and *Paramecium*?

Brain and behavior

Like the single-celled organisms the worm retreats from noxious chemicals, but its decision is more finely judged. A single sensor, the neuron labeled ASH in the brain's wiring diagram, controls this behavior by driving a "retreat" command interneuron, AVA, which shuts down the "forward" motor neurons and activates the "backward" motor neurons (figure 2.8). The sensor ASH expresses molecular receptors and detectors for a variety of potential threats, such as heavy metals, detergents, acids, or high temperature. Each input contributes to ASH activity, and when their sum suffices to trigger the command neuron, the worm backs off. Thus, a single neuron ASH serves as lawyer, jury, judge, and enforcer. It defines what constitutes

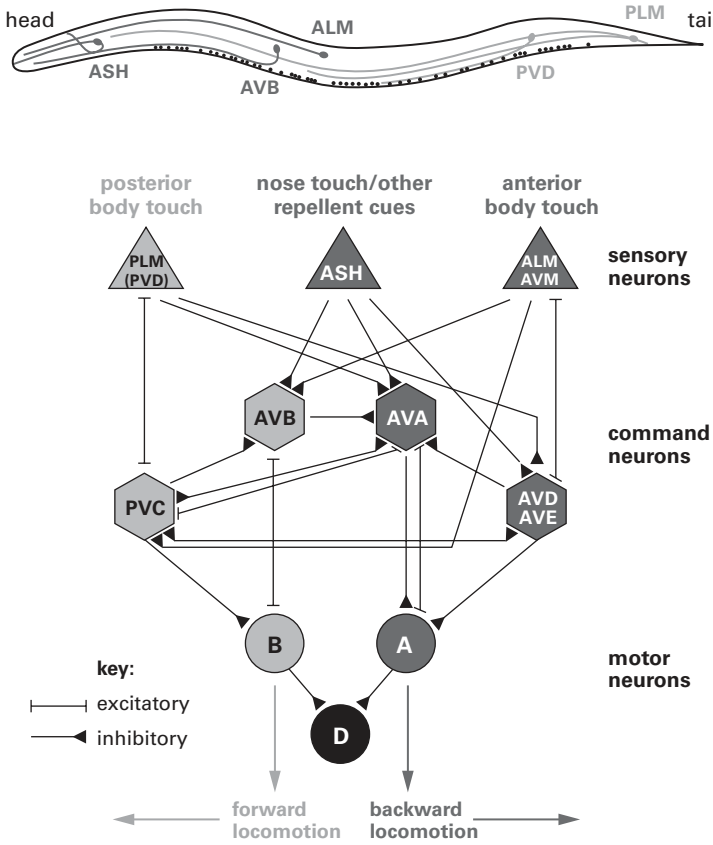


Figure 2.7
The circuit for aversive behavior. Mechanosensory neurons in the nose and in other anterior parts of the body drive command neurons for "backward" motor neurons. Mechanosensory neurons at the posterior end drive command neurons for "forward" motor neurons. These two pathways cross inhibit at the levels of command neurons and motor neurons. Adapted from de Bono & Maricq (2005), with permission.

evidence by selecting which receptors to express on its surface, collects the evidence, weighs it, judges if it warrants escape, and mandates the decision. The worm has several such sensory neurons, collecting other lines evidence for other actions.

Finding warmth, food, and mates

The worm seeks congenial places to feed, grow, and mate. *C. elegans* thrives and reproduces in a fairly narrow range of conditions: dim light,

temperature 13°–25° C, oxygen concentration 7%–14%, moderate pH, ample bacteria, and so on. To find these conditions, the worm needs a signal to warn it of imminent departure from the range—"bacteria depleted," "temperature dropping," and so forth. This search signal activates forward crawl. Foraging now for bacteria by taste and smell, the first whiff activates gradient ascent. Upon reaching favorable conditions, the worm needs a stop signal to announce "satisfaction"—what was sought is found. This signal activates a sequence of turns that places the worm in graze mode. But the worm remains vigilant. If at any moment sensors for noxious conditions are activated, they suppress the forward movement and turning, and they activate reverse.

The worm retains *E. coli*'s basic strategy for moving up or down a gradient, the biased random walk. As conditions improve, the worm turns less and runs ahead more; as conditions worsen, it turns more and runs ahead less. The mechanism is also similar: molecular receptors that drive the forward run adapt, and the decay of their output signals allows a turn. Stronger signals decay more slowly, prolonging the run.

However, with multicellularity comes an advance: ascending the gradient with paired sensors. For salt, a sensor on the right side of the head is excited by *increasing* salt, and a sensor on the left side is excited by *decreasing* salt. The right sensor excites the "forward" circuit and inhibits turning. Once the worm finds the peak concentration, this cell falls silent. If the worm moves off the peak, the left cell, excited by decreasing salt, reduces forward motion and excites turning. This search pattern, brief forward motion followed by turning, continues until the concentration starts to rise again.

The worm uses head wagging to expose both sensors to new territory and combines this action with forward thrust. This exemplifies a motor output modulated by sensing. This system also provides a case where two communication channels collect *identical* information, by sensing the same gradient, but extract different patterns and use them to drive opposite motor responses. Here is something else that a brain offers—new forms of pattern recognition that improve foraging.

Improved sensing and control are needed because *C. elegans* is to *E. coli* as a supertanker is to a rowboat. To steer a whole organism in random directions with gradual correction works on a small scale, but on a larger scale it becomes wasteful. Better for the worm to be more discriminating, to search with its *head* and inform the body once a course can be plotted. In still larger animals the sensors themselves are motorized—an insect antenna, a cat external ear, a human eye (chapter 4).

Because most worms use the same foraging circuits, they accumulate at the same sites—like undergraduates at a good café. And the subtext is similar: a place to feed is also a place to find mates. Moreover, the worms, unlike most undergraduates, are commonly hermaphroditic, so doubling their chances of a satisfying encounter. Even so, many worms enhance their attractiveness by releasing a pheromone to which intrinsically social worms are attracted. Movement toward the pheromone is controlled by a single neuron, RMG, a network hub that collects and integrates inputs from a suite of sensors and pheromones and drives the appropriate command interneurons (figure 2.8). A worm's degree of sociality is adjusted by a particular peptide released within the brain in response to changing conditions. The peptide, one member of a class of *neuromodulators*, binds to receptor proteins on specific neurons to change their activity—and hence behavior (Bargmann, 2012).

Stick and carrot

When local conditions begin to deteriorate, some definite signal is needed for the worm to move on. One such signal is the neuromodulator, octopamine. When food reserves fall, certain neurons release octopamine, which binds to receptors on particular target neurons, modifying their excitability and changing their synapses. This inhibits turning and activates the forward motor pattern. Thus, a single agent, released in response to a change in conditions, acts on specific neurons to alter circuits and switch the worm's program from "graze" to "roam."

When food is found, roaming stops and grazing resumes. This involves a second neuromodulator, dopamine. In mammalian brain, dopamine signifies (among other things) that a reward has exceeded its expected value. In worm, dopamine is released by the presence of food when, for example, mechanosensors touch particles the size of bacteria. Dopamine binds to receptors on target neurons, turning off the octopamine receptors and restoring the circuit to its previous configuration. This switches the worm from roaming to collecting its food reward. Thus, two neuromodulators, octopamine and dopamine, provide this tiny brain with a primordial stick and a primordial carrot to mediate, as they do in larger brains, "anxious" searching and "pleasurable" repetition (de Bono & Maricq, 2005).

Imminent starvation is not the only stress. Others include low oxygen, high CO_2, acidity and overcrowding. All suggest an exhausted patch—time to move on. As with humans, stress increases urgency. A comfortable worm

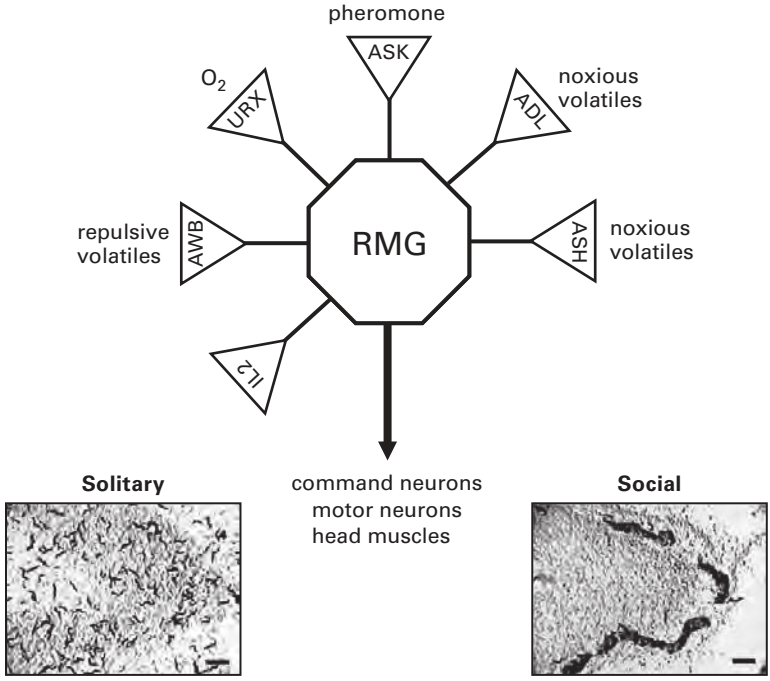


Figure 2.8

C. elegans. **Spoke and hub circuit controls solitary versus social behavior (dispersed vs. huddled). Upper**: The neuron RMG integrates social cues sensed by particular sensory neurons, ASK etc., and drives neurons that implement behavior. **Lower**: Social behavior. Solitary worms disperse and keep apart. Social worms huddle in groups. Each worm appears as a dark speck. Diagram adapted from Sokolowski (2010). Solitary and social worms from de Bono & Bargmann (1999), with permission.

moves leisurely up a promising chemical gradient, but a worm subjected to low oxygen for several hours ascends quickly. To change from stroll to rush, neuromodulators reconfigure the circuit for gradient ascent (Bargmann, 2012). For example, the sensors ADF and ASG respond to low oxygen by releasing another neuromodulator, serotonin.

Just as "carrot and stick" oversimplify human motivation, so it is for the worm. Competing for limited resources requires many factors to be weighed in deciding whether to roam or graze. A rich suite of neuromodulators allows the worm's brain of 302 neurons to evaluate contextual factors, such as nutritional status, food availability, crowding, and social signals, and then reconfigure accordingly.

Associative learning and memory

When life is good, the worm completes its life cycle (egg to egg) in 3.5 days and lives for several weeks. With a life span extending beyond the next mitotic cycle, allowing a past and a future, it now pays to recall what was good and what was bad. Far from living in the moment like *E. coli*, the worm uses its brain to associate events over time and thus draw on its experience (Ardiel & Rankin, 2010).

A worm remembers the temperature at which it was well fed and later seeks this temperature by moving up or down a thermal gradient. Finding the preferred temperature, it hangs there, searching along the isotherm. But dopamine decays promptly, so if the cupboard is bare, preference turns to aversion and the worm crawls off. Upon finding food and thus earning another shot of dopamine, the worm resets its temperature preference.

The mechanism for this learning resides within the thermal sensor that drives oriented crawling. This neuron senses changes of 0.003° C. Its response is minimal at the preferred temperature and rises on either side. The temperature for this minimum is reset by adjustments to the neuron's internal signaling; this requires protein synthesis and takes several hours. This learning process—chemical reprogramming within a single neuron—changes protein molecules but not synaptic connections.

Chemical preferences can also become associated with particular signals. For example, NaCl (salt) normally attracts worms, but when a worm has been starved in the presence of salt for only 10 minutes, it later avoids salt. A particular neuron downstream from the salt sensor releases another neuromodulator (insulin) that feeds back to an insulin receptor on the salt sensor to activate an internal signaling pathway (involving PIP3-kinase) to suppress attraction. Again, reprogramming a signaling pathway *within* a neuron allows experience to change the balance between attraction and repulsion. This mechanism also serves odorants. *C. elegans* even learns to avoid odorants from a particular pathogenic strain of bacteria that has made it sick.

These memory traces promote survival by extending the time over which an animal can identify and use patterns. The number of trials needed to establish an association is modest, five to ten repeats over 20 minutes. This makes sense in an environment where conditions are sufficiently shifty that to be useful, an association must establish rapidly and decay rapidly. In short, the worm's behavior demonstrates its reliance on information from three distinct sources: outside, inside, and the past. Its brain integrates these streams to select behaviors that, reflecting a wider context, improve the worm's vitality and reproductive success.

Some design aspects of this tiny brain

C. elegans' brain may be small, but it is not simple. To achieve its panoply of behaviors, the worm draws on a large catalog of molecular parts. This includes diverse proteins for intracellular chemical and electrical signaling, plus numerous parts for processing information at synapses. For example, signaling proteins occupy 20% of the worm's genome, and its 300+ synaptic parts amount to one third the number for mammals (Emes et al., 2008). In fact the worm brain uses many of the same components present in larger brains. Since parts are shared, one might expect some design rules to be shared as well. If some rules were not shared, that would also be instructive, for it might suggest costs and benefits of scaling up.

Here then are some design features gleaned from considering the worm's brain and what they might imply for bigger brains.

Computes as much as possible within a single cell

This feature is exemplified by the worm's *receptors* and their *sensors*. We distinguish these terms: "receptor" refers to an individual *protein molecule* that responds to a specific event—like stretch, temperature, protons, or chemical binding; "sensor" refers to an individual *neuron* that expresses one or more types of receptor. Although neuroscientists understand this difference perfectly well, for historical reasons they often use "receptor" for both the molecule and the neuron. We use different terms to reduce confusion for readers unfamiliar with the jargon, and also because they raise two design problems.

First, a single receptor molecule is subject to stochastic fluctuations, such as thermal noise. Therefore a neuron might need to improve the signal-to-noise ratio of signals conveyed by one receptor by averaging over a population of the same type. This raises the following design question: How many receptors of the same type should be expressed by each sensor? The answer will be given in chapter 6.

Second, receptors are more diverse than the sensor neurons that express them. Therefore, how should diverse receptor types be apportioned among sensors? For this problem *C. elegans* has a rule. If a set of receptors all lead to the same final action, they share a common sensor. For example, the sensor ASH collects signals from various types of receptor for noxious stimuli that require an aversive response; ASH couples its output to a single neuron that executes a command: *Scram!*

This rule explains receptor grouping generally. The worm uses more than 1,700 different types of receptor molecule for chemoreception (taste

and olfaction). This considerably exceeds the 800 or so used in mammals, but unlike mammals where each receptor type is typically assigned its own sensor, the worm provides only about 30 separate sensor neurons. Like sensors of noxious stimuli, each chemosensor sends its signal to a specific command neuron. So the signals from 1,700 different input channels (receptors for taste and olfaction) are assembled for action, not by circuits higher in the brain, but by a few dozen sensory neurons.

Computing *within* a cell economizes on neuron numbers. The worm meets all basic requirements for behavior (sensory pattern recognition, sensorimotor integration, and motor control) with small numbers of neurons. Thirty-eight sensors connect to 82 interneurons (whose processes are confined within the brain) that contact 119 motor neurons (cells whose processes leave the brain to contact the worm's 100 muscle cells). This reserves about 70 neurons for internal regulation and mating.

Yet there is a downside to performing several operations in a single cell. A cell's capacity to handle information is limited by factors such as internal noise, dynamic range, and energy supply. So a sensor that processes inputs from several types of receptor compromises its ability to handle the information from any one receptor type. A dedicated sensor can devote more receptors to its particular modality and thus improve sensitivity and signal-to-noise. This is the engineer's principle from chapter 1: to prevent one component from doing two tasks suboptimally, complicate.

Complication goes up the line. Better sensors warrant better sense organs: eyes for vision, ears for hearing, and so on. To benefit from these more accurate and discriminating sense organs, specialized sensory systems evolve in larger brains, each devoted to processing a single modality. The conclusion is obvious: as brains scale up to improve behavior, neurons specialize. Chapter 3 will suggest how and why, but now we consider a related question, how does a worm's tiny neuron manage to compute efficiently?

Uses chemistry wherever possible

Many worm neurons use internal molecular circuits to perform functions that in larger brains use a circuit of several neurons. For example, a single sensory neuron, AFD, determines the worm's temperature preference by adding new proteins to its intracellular signaling network. Another neuron, AWC^{ON}, changes a behavioral response to suit the situation. When an odorant is present *without* food, AWC^{ON}'s molecular receptors adapt and chemotaxis declines. However, when the same odorant is present *with* food, its receptors are sensitized, and chemotaxis increases (de Bono & Maricq, 2005). These competing responses are controlled by an intracellular

mechanism that switches the connection between sensor and behavioral output to reverse the control of chemotactic turning behavior (Pereira & van der Kooy, 2012).

These examples show that chemical computing by circuits *within* a neuron can manage behavior. Moreover, this can be very efficient because chemical signals are orders of magnitude cheaper than electrical signals (chapters 5 and 6). Chemical diffusion is slow for long distances, but the worm *is* small and slow. Thus, the worm's size and speed well suit its reliance on cheap chemical signaling. In addition, chemical signals can be broadcast to specific targets, which brings us to another design feature.

Uses neuromodulators to switch behaviors

Three neuromodulators were mentioned (octopamine, serotonin, and dopamine) that switch the worm's behavior in response to stress or the prospect of reward. But this is just page one from the parts catalog since the worm expresses 250 small peptides with known neuromodulatory functions. Their diversity and ubiquity is understandable because neuromodulation is so ingenious (Harris-Warrick & Marder, 1991). A neuromodulator can be broadcast widely yet still act locally and specifically, affecting only neurons that express an appropriate receptor. The receptors often couple to a protein that modulates intracellular signaling, so in effect a neuromodulator uses *trans*cellular chemistry to modulate *intra*cellular chemistry.

A neuromodulator's reach is further enhanced because its receptor diversifies into multiple subtypes that couple to different intracellular signaling networks. Consequently, one small molecule can retune and reconfigure a whole neural circuit without altering the anatomical connections. This allows every circuit to always be doing something and then to be recruited for something else as required. Thus, neuromodulators allow the brain to use components to their fullest.

Conserves synapses

The worm brain makes only about 6,400 chemical synapses. This is roughly the number that in a mammal contact a single retinal ganglion cell or a single cortical pyramidal cell. How can a worm operate with so few synapses? The neurons are far smaller and therefore can be driven by fewer synapses. But since a single synapse is unreliable, how can so few synapses signal reliably?

One answer is: *slowly*—a neuron can improve reliability by averaging over time. This can be tolerated because, compared to many animals, the worm lives in the slow lane. For example, its olfactory sensor uses a

chemical amplifier, a G protein signaling cascade that integrates for more than 20 s (chapter 5, figure 5.6). This sensor drives a synapse that integrates over several minutes. By comparison, a fly's olfactory system acts in less than 1 s. Locomotor waves descend the worm's body at 1 Hz, but an insect moves its legs faster than 10 Hz. So the worm can prosper with few synapses because it is slow. This suggests another feature: *send information as slowly as possible* because this uses fewer synapses, smaller cells, and less energy. Later chapters explain more.

Uses stereotyped components

Efficient design gives every component a definite task. Once all components are optimized for their tasks and optimally fitted together, it is efficient to repeat them across individuals. Similarly, every neuron in *C. elegans* has a definite role optimized by natural selection to meet a specified level of performance. Correspondingly every neuron is "identified," meaning that it exhibits a stereotyped morphology, chemistry, and location in every animal (White et al., 1986). The circuits are also identified, meaning that the synaptic connections are essentially identical across animals. This was established by reconstructing the entire nervous system from thousands of electron micrographs of serial sections—to produce the worm's *connectome* (figure 2.9). Identified neurons and circuits are consistently found in small brains: worm and water flea, leech and lobster, and so on.

Minimizes wiring costs

The layout of *C. elegans*' neural wiring suggests that all 302 neurons are located as near as possible to the sites where they are needed (Varshney et al., 2011). Chemical and thermal sensors concentrate at the head; tactile sensors that guide locomotion distribute along the body axis; motor neurons that propel the worm forward distribute along the rear half of the body, and motor neurons for reverse locomotion distribute along the front half (figure 2.6). But does the layout approach the optimum sought by chip designers—the unique set of placements that minimizes the total length of connections in the brain?

Designers of silicon chips have developed algorithms to optimize component placement. Their rule: place the most densely interconnected components close together and the more sparsely connected components further apart (figure 2.9). This algorithm applied to the worm's brain shows that 90% of neurons are optimally positioned (Cherniak, 1995; Chen et al., 2006; Pérez-Escudero & de Polavieja, 2007). The 10% of neurons not in their optimal position suggests competing needs. For example, neurons

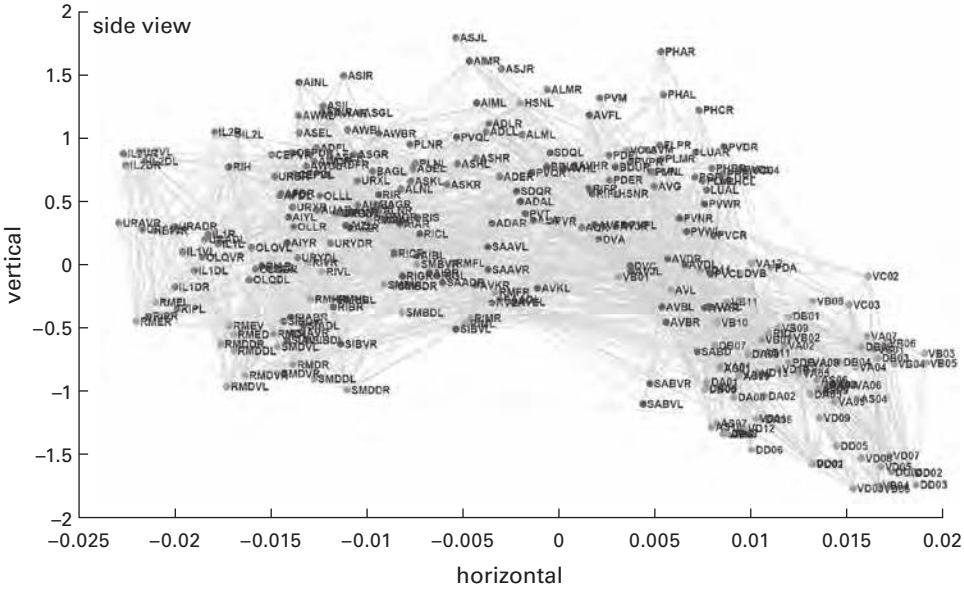


Figure 2.9

***C. elegans* connectome reconstructed from serial sections photographed in the electron microscope.** Each neuron is identified, and its synaptic connections are shown in gray. At the time of writing this is one of the most complete wiring diagrams established for any part of any brain (the other is the fly lamina cartridge, figures 9.2 and 9.3). Careful estimates suggest that this worm connectome is 93% accurate. Such are the technical difficulties of tracing neurons' thin connections that, after two decades of work on 302 neurons, 7% of connections are "missing." Reprinted with permission from Varshney et al. (2011).

that communicate most frequently with each other may be placed closer together to save energy and reduce conduction delays between them. Although layouts in larger brains certainly reflect this, conduction delay may be less relevant for *C. elegans* because the distances are so short, and the worm is so slow. "Short and slow" suggests another design feature.

Favors analogue over pulsatile

Because electrical signals in the worm travel less than a millimeter, neurons can conduct passively, as graded (analogue) changes in electrical potential. The brief, sharp, energy-intensive action potentials that dominate long-distance signaling in larger brains are unneeded, so the worm can rely solely on analogue computations, which are direct and energy efficient

(Sarpeshkar, 1998). Even its motor neurons operate in analogue mode. Over these short distances, analogue signaling transmits more information per neuron and at lower cost (chapter 7). So firmly does *C. elegans* hold to this feature that it has abandoned the gene that encodes the voltage-gated sodium channel used by larger, faster species to produce spikes.

Conclusions

Three organisms of ascending size, *E. coli*, *Paramecium*, and *C. elegans*, show why an animal needs a brain to process information on a larger scale. It is to increase opportunities for survival and reproduction in a competitive and variable environment.

The small single cell, *E. coli*, survives with surface receptors that relay information to the internal chemical signaling networks that determine metabolism, growth, reproduction, and movement. However, *E. coli* is a mere speck in space and time with most opportunities beyond its reach. A larger cell, *Paramecium*, moving more briskly travels farther, expanding opportunities, but is ultimately limited by its chemical signaling networks—diffusion and internal communication by intracellular motors are both too slow. Voltage-gated ion channels added to the cell membrane allow fast electrical signaling, but trapped in a viscous world, a single cell can only do so much.

The multicellular worm, *C. elegans*, overcomes viscosity by enlarging, and it moves faster and farther by specializing cells. This leads it to more opportunities and dangers—richer sources of information to be gathered and processed that finally need a brain. The key innovation is the neuron, a cell type specialized to collect, process, and communicate. Each neuron links its rich web of internal chemical communication to the electrical network at the surface membrane and thence to other neurons via synapses. Neuromodulators retune selected neurons to reconfigure whole circuits. Thus, a brain of only 302 neurons extends the worm's horizon by providing a behavioral repertoire that adapts to changing contexts.

The worm accomplishes the same tasks as a bacterium or protozoan—finds growth conditions and mates while avoiding unproductive or toxic sites. And it does so with similar behaviors, such as gradient ascent by biased random walk and avoidance. But with its brain *C. elegans* can cover more territory, and with its longer lifespan (weeks instead of minutes), it can adapt to nasty surprises as an *individual* rather than as a miniscule part of an adapting *population*.

Here emerges another design principle. Life span and lifestyle are related to the appearance of particular types of memory and particular decay times. Nothing should be remembered that is unlikely to enhance survival and reproduction. Nor should memories exceed the typical time constants of useful correlations—because when correlations decay, memory ceases to predict anything useful. But it *is* useful to establish the memory trace rapidly before it is outdated—and that seems to occur—few trials, closely spaced. This suggests that the longest and deepest human memories are not mere decoration but serve to shape character over a lifetime, promoting survival in our complex social fabric (chapter 14).

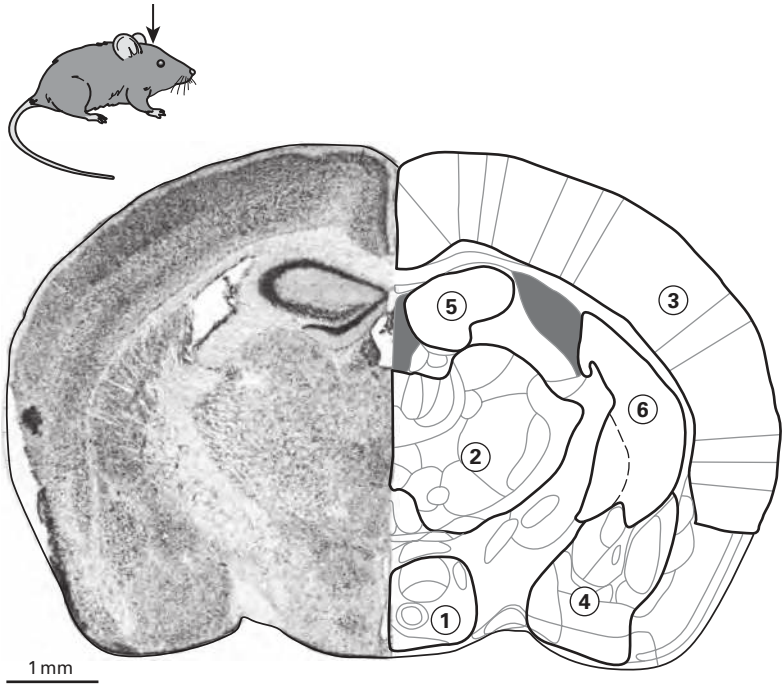
Finally, given that *C. elegans* does so well with only 302 neurons, one might look critically at an assumed truth—that it is better to have a bigger brain. So why *have* animals evolved still bigger brains?

3 Why a Bigger Brain?

This chapter will explain why, despite the worm's success with 302 neurons, brains expand. The mouse cerebral cortex contains about 10^7 neurons. This seems like a lot until you consider that the cortex of the macaque monkey, a key experimental model, is larger by 100-fold, and that human cortex is 10-fold larger still (Herculano-Houzel, 2011). Despite this huge range of scales, one feels comfortable generalizing about the "mammalian brain"—because every part identified in mouse can also be identified in macaque and human (figure 3.1; Kaas, 2005).

Consider also the fly brain. It has 500-fold fewer neurons than the mouse brain, but 500-fold more neurons than the worm brain, plus a rich structure—so warranting a slot in the "large brain" category. Insect and mammal brains share many similarities. For example, both gather their neurons into clusters and their axons into cables (*tracts*). Both employ special structures to accomplish the same broad tasks: store high-level input patterns, generate low-level output patterns, and retrieve patterns using reduced instructions. Of course, there are differences, given the differences in body design and behavior. Yet, despite half a billion years of evolutionary opportunity to diverge, brain designs in insect and mammal seem to have followed the same rules.

For designs to have persisted across this immensity of time and spatial scale implies that they are neither arbitrary nor accidental. Rather, they must have emerged as responses to some broad constraint. That is what elevates the shared responses to the status of *principles*. This chapter will identify the key constraint and indicate how it leads to three principles that govern the organization of larger brains.



1. Generate patterns for wireless signaling and appetitive behaviors.
2. "Preprocessing" to shape signals for higher processing.
3. High-level processing: assemble larger patterns, choose behaviors.
4. "Tag" high-level patterns for emotional significance.
5. Store and recall.
6. Evaluate reward predictions.

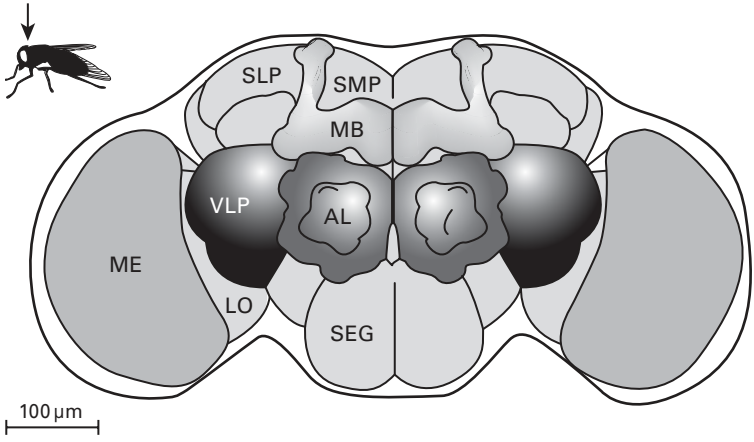


Figure 3.1
Mammalian and insect brains share many broad aspects of design. **Upper**: Cross section through mouse brain; inset indicates plane of section. **Left**: Fine dots are neurons; dark regions are neuron clusters; bright regions are myelinated tracts (chapter 4). **Right**: Numbered regions dedicated to core tasks: (1) *hypothalamus*; (2) *thalamus*; (3) *cerebral cortex*; (4) *amygdaloid complex*; (5) *hippocampus*; (6) *striatum*. Reprinted with modifications and permission from Franklin and Paxinos (1996). **Lower**: Cross section through fly brain; inset indicates plane of section. Brain is built of more than fifty clusters, each specialized for particular tasks. Depicted here are ME, medulla—detect and map local visual patterns; LO, lobula—assemble local visual patterns into larger patterns; AL, antennal lobe—preprocess olfactory signals for pattern recognition; VLP, ventrolateral protocerebrum; SLP, superior lateral protocerebrum; SMP, superior medial protocerebrum—all involved in high-level integration; MB, mushroom body—store and recall; SEG, subesophageal ganglion—integrate information for wired and wireless output to body.

A brain's core tasks

As animals emerge from the soil to a wider, less viscous world, the possibilities for foraging expand immensely. A worm explores mainly in two dimensions over an area of 0.01 m^2 whereas a honeybee typically covers an area of nearly 10^7 m^2, and a fly somewhat less. So foraging area expands by 10^9 (1 billionfold). Add the third dimension, and the volume to be explored becomes astronomical. Larger animals, such as fish, birds, and mammals, may migrate and thus forage over thousands of kilometers—thus millions of square kilometers.

Such gigantic territories contain immense resources and, of course, harbor innumerable dangers. For an animal to find the one and avoid the other requires it to rapidly gather vast amounts of information from the environment. To calibrate "vast" with one example, the eye sends the brain about 10 megabits per second, roughly the rate of an Ethernet connection (Koch et al., 2006). All sense data reach the brain in the form of tiny patterns—evanescent pieces of a dynamic jigsaw puzzle—and to be of any use, they require assembly to reveal a larger pattern. So if gathering information is to be at all rewarding, the brain must commit resources to assembling larger patterns on spatial and temporal scales that are relevant to behavior.

Yet, even a larger pattern might be useless until it is compared to a library of stored patterns where it can be identified: *edible/toxic*, *friend/foe*, or *search item not found*. Either outcome provides a basis for behavioral choice. A

match allows confident choice: eat or decline, approach or flee. A non-match suggests caution and need to gather more data. Thus, the brain requires "pattern comparators," and these must couple to mechanisms that select behaviors: *feed*, *fight*, *copulate*, *investigate*. These, in turn, couple to mechanisms for detailed motor patterns to drive muscles for moving limbs or wings.

Any given motor behavior *might* match exactly the action that was ordered: the arrow might strike the exact point at which it was aimed. But often there are errors due to environmental or neural perturbations, and these need to be identified, so that performance can progressively improve. Thus, a brain needs mechanisms to evaluate the mismatch between the orders it gave and the actual motor performance. So, in addition to sensing and processing patterns to discover "what's important out there," the brain also devotes considerable resources to sensing and processing its own motor errors, and other errors of internal "intentional" signaling in order to improve the accuracy and efficiency of the next round. This is "motor learning."

Behaviors are subject to another important class of errors. Every action has both costs and consequences. The costs are partly energetic: how much energy was spent? But also there are "opportunity costs": could the return have been greater and the risk less for some different action? Every behavior, even when perfectly executed, needs to be evaluated from this perspective: wise or foolish? repeat or not? These evaluations of *reward prediction*, like those for motor errors, are used to update stored knowledge in order to improve the next round of predictions. The nematode worm already shows this type of evaluation to some degree, but animals in the wider world allot it major neural resources.

In sum, to succeed in the wider world, an animal must exchange larger amounts of information with its external environment and also evaluate the costs and consequences of its actions. The seven core tasks that every brain must accomplish are summarized in figure 3.2. What the brain does for the external environment it also does for the internal environment which has also expanded and complexified. Moreover, the mechanisms for managing the internal and external environments need to couple closely in order to serve each other (figure 3.2).

Why the internal milieu needs a brain

To support richer external behaviors, an animal requires specialized internal tissues and organs. Some digest the bounty foraged from the outer

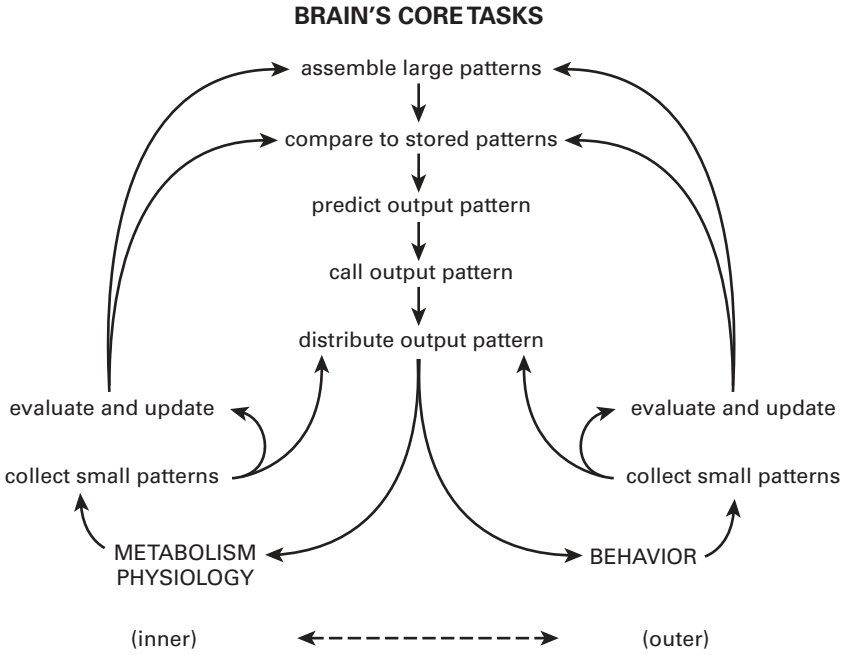


Figure 3.2
Large brains accomplish the same broad tasks. Note that inner and outer tasks couple to serve *each other* (↔).

world; others store metabolites and energy-rich compounds for release upon demand. Still others regulate ionic balance and cleanse the internal milieu, or distribute oxygen and metabolites to hungry tissues. Specialized organs of immunity protect against infectious agents and parasites. Organs couple to form systems, and systems cross-couple to optimize overall function.

The standard idea is that the internal systems more or less take care of themselves. Each parameter is supposed to have a set point, like a thermostat, from which deviations trigger feedback to correct the mismatch (*homeostasis*). Internal regulation also employs *autonomic nerves*—so termed because they are in some sense independent of voluntary control—thus, autonomous. We cannot "will" our heart to beat faster or our blood pressure to decrease. However, we can accomplish these shifts by recalling or imagining the appropriate scene. This implies the existence of neural pathways from pattern stores to pattern generators for autonomic circuits. Thus, although the autonomic nerves are generally supposed to serve

emergencies ("fight or flight"), they actually serve continuous regulation—not just for panic, but for efficiency.

Efficient regulation anticipates

In fact, all internal regulation, even the mildest sort, is far from autonomous. As the external environment presents opportunity or cause for concern, internal processes must predict what the external environment is about to deliver and must prepare particular responses that will probably be needed in support. For internal processes the goal is not to correct mismatches but to prevent them.

Such predictive regulation was demonstrated for feeding and digestion by Ivan Pavlov more than a century ago: the brain processes small patterns from the outside (sight or smell of some substance) and matches them to a stored pattern that identifies a particular food. Then the brain triggers secretions all along the digestive system to prepare for what's coming, starting in the mouth (if bread, then amylase; if fat, then lipase), then on to the stomach (if meat, then acid plus protease), the intestine (if fat, then bile), and finally the circulation (if glucose, then insulin). All of these secretions occur *before* and *during* the meal, triggered *predictively*—anticipating what will be coming down the gastrointestinal tract—thus preparing systems for absorption and uptake in order to prevent deviations that would need correction by negative feedback (Fu et al., 2011).

Modern work extends this point: as the stomach releases its contents to the next stage, it also signals the brain to prepare for the next bout of foraging. The brain responds by tuning up sensitivity of the olfactory receptors and by increasing the rate of sniffing (Julliard et al., 2007; Tong et al., 2011). Thus, the stomach warns the brain "Prepare to forage again"—well before the body has begun to deplete its reserves. Moreover, as fat reaches the small intestine, the gut can predict confidently the approach of satiety. Therefore, the gut warns the brain "cease feeding and proceed to the next activity"[1] (Fry et al., 2007).

Each "next activity" requires the brain to predict continuously, and in timely fashion, the need for a particular blood pressure. Consider the record of mean arterial pressure over 24 hours (figure 3.3). In early afternoon, as the subject attends a lecture, his brain anticipates reduced demand and allows him to doze: pressure falls. Startled awake by the jab of a pin, the brain predicts danger: pressure spikes; then, identifying a prank, the brain directs the nap to resume: pressure falls. At midnight the subject has sexual intercourse: pressure spikes, but then falls profoundly and stays low during sleep. Come morning, the brain predicting a busy day, restores the pressure.

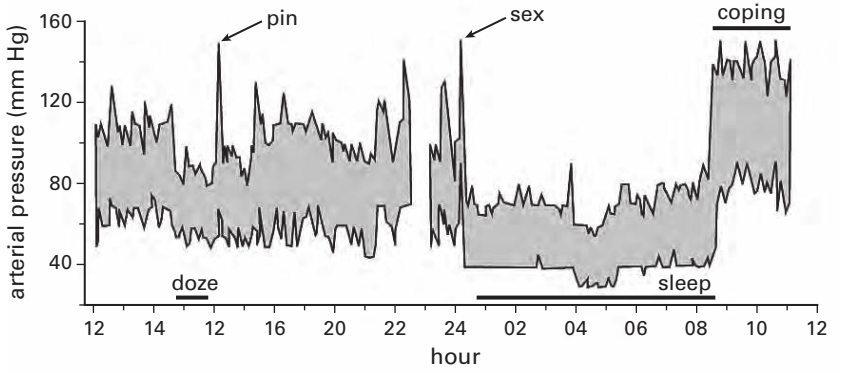


Figure 3.3

Internal systems match behavior. Arterial pressure fluctuates with demand. Each shift in pressure is accompanied by parallel shifts in hormonal and neural signaling that follow the broad catabolic and anabolic patterns. Redrawn from Bevan et al. (1969) and reprinted from Sterling (2004b).

Such anticipatory tuning requires coordinated action of multiple organs and organ systems. To raise pressure, the heart accelerates and vessels constrict. Also the kidney expands blood volume by pumping more salt water into the circulation. The kidney also signals the brain that the body will soon need more supplies of salt and water. Thus, like the gastrointestinal tract, the kidney alerts the brain well in advance of an upcoming need to resupply. Each contribution operates on a different timescale: faster for heart and vessels, slower for kidney's pumping, and still slower for the brain's rise of salt appetite and thirst. These contributions to internal regulation are all initiated simultaneously—and largely by the same signals.

In short, every move we make is matched by a corresponding cardiovascular and renal pattern. Of this we are generally unaware. Yet if the motor command ("Arise!") slightly precedes the internal command ("Tighten vessels!"), blood flow to the head drops, and we faint. That this experience, *postural hypotension*, occurs rarely attests to the rigorous coupling between the cardiovascular pattern and muscular patterns on a 100-ms timescale. On a slower timescale "Arise!" increases by eightfold a signal to the kidney to save water.[2]

Note that matching blood pressure to environmental context requires all of the brain's broad tasks as diagramed in figure 3.2—the collecting and assembling of patterns, the comparison to stores, and so forth. How else to decide if the jab is from a friend or enemy? Moreover, every high-level call

to external action is delivered simultaneously to multiple internal organs. Thus, collecting patterns and distributing patterns are both thoroughly coupled between inner and outer worlds. Where and how the brain effects this coupling will be treated in chapter 4.

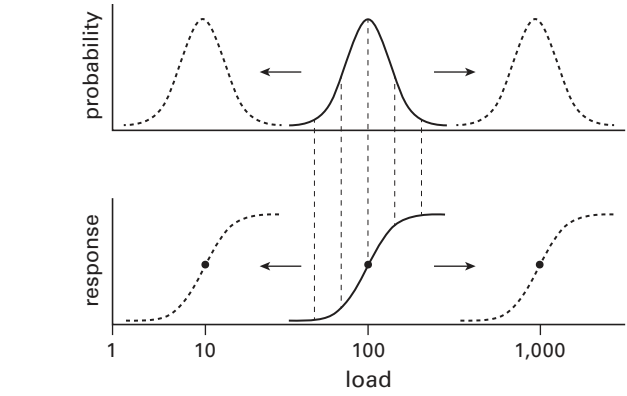
Adapt, match, trade

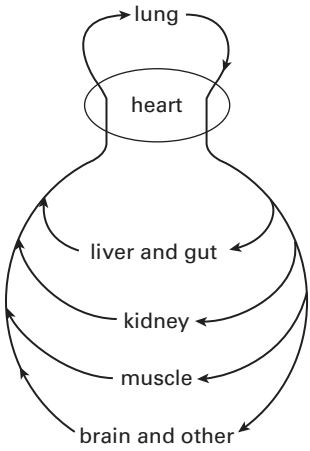
Although this book concerns efficient neural design, we must keep in mind that the brain comprises only 2% of the body's mass and 20% of its energy. So the body also needs to operate efficiently. Each organ should match its capacity to the anticipated need of the organ downstream. Too little and the system will fail; too much and capacity is wasted. So each organ needs constant tuning to anticipate the next demand (figure 3.4). But what happens when a need exceeds the capacity to supply? This problem is solved by arranging various short-term "trade-offs." Such cooperation enhances the range of performance while greatly reducing average excess capacity (figure 3.4).

For example, the "resting" heart pumps 6 L of blood per minute through the respiratory system and then out to the general circulation. Resting skeletal muscle uses about 20% of the oxygenated blood—matched to its modest need for maintaining posture. During peak exercise, muscle must increase its supply by nearly 20-fold, but the pulmonary and systemic circulation can increase their outputs only fourfold. Therefore, the body must either reduce its peak capacity for exercise or increase its peak pulmonary and cardiovascular capacity by fivefold—imagine the chest! Or it can borrow.

Indeed, during peak exercise the splanchnic circulation (gut and liver) and the renal circulation (kidney) both reduce their shares by four- to fivefold, enough to pay part of muscle's bill for exercise. During digestion, when the splanchnic circulation needs more blood, it borrows from muscle and skin—unless skin needs blood for cooling. The brain neither makes loans nor allows overdrafts that might cause it to overheat. Anyone who has eaten and then exercised in the sun will recall how these conflicting demands from muscle, gut, and skin are resolved: by corrective motor commands to internal systems ("Vomit!") and to external systems ("Lie down!"). Moreover, the experience receives a strongly negative evaluation that updates the knowledge store ("Do not repeat!").

This example illustrates three key rules for efficient regulation: (1) adapt response capacity to changes in input level, (2) match response capacities across the system, and (3) trade between systems. Regulatory responses begin promptly—as soon as there is sufficient statistical evidence to predict





Blood Flow (liters per minute)

	Rest	Exercise	Change
lung, heart	6.0	25.0	+ 325%
liver, gut	1.5	0.3	− 80%
kidney	1.1	0.3	− 80%
muscle	1.2	22.0	+ 1800%
brain, other	2.2	2.5	+ 10%

Figure 3.4

Adapt, match, trade. Upper: Adapt response capacity to load. Every system confronts some distribution of probable loads (bold). As conditions shift, so does the distribution (dashed). The response curve (bold) is typically sigmoid with its most sensitive region (steep part) matched to the most probable loads. As a sensor detects a statistically reliable change in the distribution, it prepares the effectors by shifting their response curves to match the new distribution (dashed). Each sensor also adapts its own sensitivity. Reprinted from Sterling (2004b). **Lower:** Organs and organ systems couple efficiently by matching loads to capacities. Trade-offs allow better performance while reducing unused capacity and enhancing "portability." Blood flow pattern changes with exercise: total flow quadruples, but that is insufficient for muscle. To meet the full need, blood is routed from liver, gut, and kidney, temporarily reducing their performance but eventually benefiting from what the muscular effort has accomplished. Data from Weibel (2000).

a new target level. By comparison, self-regulation by feedback to a set point would be hopelessly inefficient. But to execute these principles of predictive regulation requires an organ with knowledge of the outside, knowledge of the inside, and knowledge of the past to anticipate what the whole animal will need over various timescales—the whole brain (Sterling, 2012).

Bigger brains

We seem to have answered "Why a bigger brain?" In a wider world, a more effective brain expands the possibilities for behavior. Control of appendages such as fins, wings, and legs lends speed and scope to exploration, so that vastly more small patterns are encountered which then require selection and assembly. More large patterns require more comparisons, requiring a larger library; more comparisons also require more decisions, and these require more evaluation. Naturally, more neurons are needed, and since neuronal components are irreducibly small (chapter 7), a brain must enlarge.[3]

The larger brain, to be effective, must operate in real time. One need not watch a sloth for very long to realize the limits to life in slow motion. The larger, faster brain must still remain portable and also metabolically affordable. So a brain needs to be both functionally effective and cost-effective. These demands for speed, portability, and affordability all interact; therefore, individually and together they raise questions of brain design. We turn now to the fundamental constraint on any brain design that leads to the first three design principles. Then, in the context of these few principles, we discuss some actual designs (mammal and insect).

Design constraints

The fundamental constraint on brain design emerges from a law of physics. This law governs the costs of capturing, sending, and storing *information*. This law, embodied in a family of equations developed by Claude Shannon, applies equally to a telephone line and a neural cable, equally to a silicon circuit and a neural circuit. This law constrains neural design at all scales and cannot be avoided any more than a B-29 bomber can avoid the law of gravity. But, though the brain is fundamentally an organ that manipulates information, few neuroscientists are familiar with this law or aware of its value for understanding brain organization. We explain it briefly here and give more detail in chapters 5 and 6.

What *is* "information"?

Information is *the reduction of uncertainty about some situation X associated with observing any variable Y that is causally correlated with X*. Uncertainty defines the standard measure: one *bit* is the information needed to decide between two equally likely alternatives. Information depends on causality because, to reduce uncertainty, a message must be reliably relatable to its source, the event that caused it. Any factor that reduces the reliability of this connection, such as noise, increases uncertainty and destroys information.

Reduction of uncertainty succinctly describes the brain's purpose. A spike in an ON ganglion cell reduces the brain's uncertainty that a brighter than average object is located in a particular region of the visual field (chapter 11). And when the brain matches the sensory pattern coded by a patch of ganglion cells to a stored pattern, it reduces a key uncertainty: "Friend or foe?" The answer helps to select the next behavior and implement it. To this end, a motor neuron spike decreases the uncertainty that its target muscle fibers will contract and help the animal move in the appropriate direction. In short, to achieve its core purpose, the brain uses physical devices (neurons and circuits) that represent and manipulate information. So now we must ask: how much information can a neuron represent, and what constrains its capacity?

A neuron's information capacity

To convey information, a neuron must represent the state of its input as a distinct output (input and output must be causally related). It follows that a neuron's capacity to convey information is limited by the number of distinctly different outputs that it can generate. The number of different outputs a spiking neuron can generate in a given time is the number of distinctly different spike trains that it can produce in that time. This depends on two factors, mean firing rate (R spikes per second) and the precision of spike timing (Δt seconds). The upper bound on firing rate is set by spike duration plus the period following a spike when a neuron is refractory (cannot spike). Certain neurons reach this limit during brief bursts, but most neurons operate far below this limit. Precision is limited by channel noise and membrane time constant. Here biophysics limits information capacity.

What is the relation between spike rate, timing precision, and the number of different spike trains a neuron can produce? When a neuron transmits for 1 s, it produces R spikes with a timing precision of Δt (Rieke et al.,

1997). The number of different spike trains, M, is the number of ways the neuron can place its R spikes in $T = 1/\Delta t$ intervals (figure 3.5). Deriving M is a standard exercise in calculating combinations that is often set to students in quaint terms, such as placing peas in pots. The solution is

$$M = T!/(R!(T - R)!), \tag{3.1}$$

where ! denotes factorial and $(T - R)$ is the number of empty (spikeless) intervals.

The number of different messages, M, that a neuron can generate in 1 s converts to information rate. According to Shannon, the information, H, is given by

$$H = \log_2(M). \tag{3.2}$$

Substituting for M using (3.1) gives

$$H = \log_2(T!/(R!(T - R(!)) = \log_2(T!)—\log_2(R!)—\log_2((T - R)!). \tag{3.3}$$

Because Shannon used a logarithmic scale, a message lasting twice as long conveys twice as much information. And, because he used log base 2, information is in bits. Thus, H, the information that a neuron can transmit with messages 1 s long, is its information capacity in bits per second (figure 3.5).

With this expression we can "follow the money." That is, using a standard currency (bits) we can ask like good engineers: how fast does a neuron send information (bits per second) and how efficiently (bits per spike)? And at what cost in space (bits per cubic millimeter) and energy (bits per molecule of adenosine tri-phosphate)? This molecule, abbreviated *ATP*, is the standard intracellular molecule for transferring energy.

Information costs energy and space

Information rate increases with spike rate and with spike timing precision, that is, reduction in Δt. However, for any given precision, information rate increases sublinearly with spike rate (figure 3.5). Consequently, as spike rate rises, bits per spike should fall, and this theoretical decline in bits per spike is observed experimentally (figure 3.5).

There is another way to explain why more frequent spikes carry less information. A symbol that occurs less frequently is more surprising and so more informative (chapter 4, equation 4.2). This effect, which Shannon called *surprisal*, makes a code with fewer spikes more efficient. For example, a code that distributes spikes sparsely among a population of neurons conveys more bits per spike (chapter 12; Levy & Baxter, 1996).

This simple law—infrequent spikes carry more bits—profoundly influences neural design because, following the money, one finds that spikes are

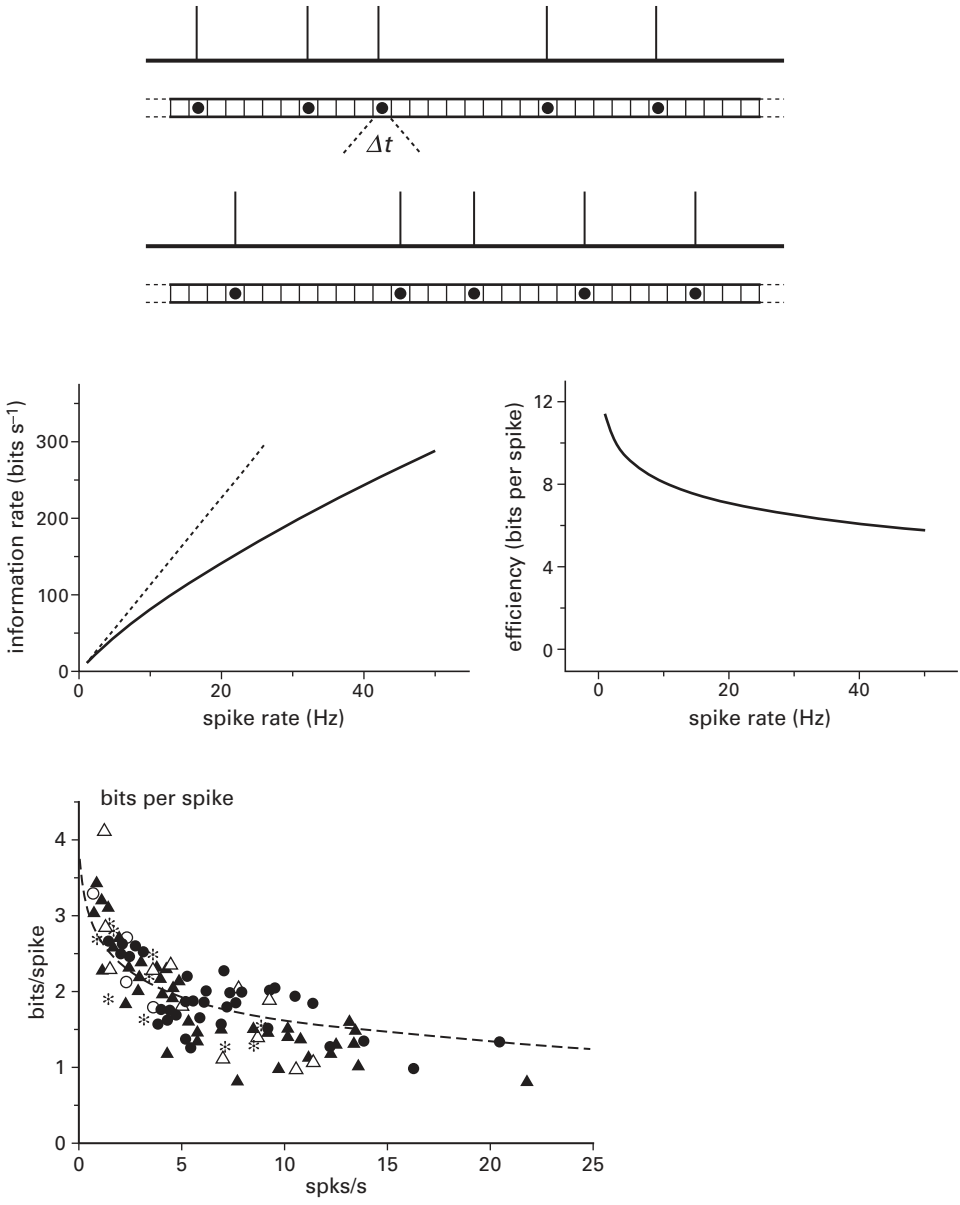


Figure 3.5

Mathematics and biophysics govern the representational capacity of signal trains. **Upper**: Distinct sequences of spikes in time intervals Δt represent different inputs. **Middle left**: Theory predicts information rate to increase sublinearly with spike rate, with the consequence shown at **middle right**: Increasing spike rate reduces the information transmitted per spike. These theoretical curves were calculated using the standard approximation for signal entropy at low spike rates (Rieke et al., 1997, equation 3.22). In general neurons do not achieve their theoretical capacity because of noise and redundancy; consequently, measured values of bits/spike are lower (figure 11.25). **Lower**: Measured bits per spike falls as mean spike rate increases. Data pooled from several classes of guinea pig retinal ganglion cell. Reprinted with permission from Balasubramanian & Sterling (2009).

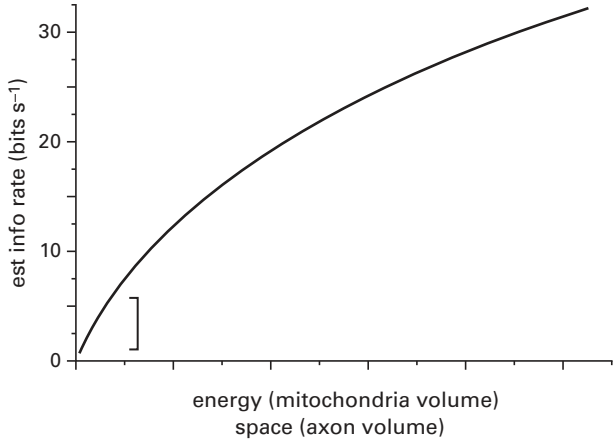


Figure 3.6
Law of diminishing returns. Doubling information rate of retinal ganglion cells more than doubles space and energy costs. Consequently, neural designs try to stay on the steep region of this empirically measured curve. Modified from Balasubramanian & Sterling (2009) and reprinted with permission.

expensive. They use about 20% of the brain's energy (Attwell & Laughlin, 2001; Sengupta et al., 2010). A spike charges a neuron's membrane capacitance by about 100 mV, and the membrane area is substantial due to a neuron's local branching. Higher mean spike rates require a larger cell body with greater membrane area; this increases energy cost per spike and adds to the cost of transmitting bits at high rates. Consequently, where spikes are sent sporadically and at low mean rates, more information can be sent for the same energy—more bits per ATP. This saving in energy by low rates is compounded by a saving in space.

Higher spike rates also require thicker axons.[4] Because axon diameter, d, increases directly with firing rate, axon volume rises as d^2; therefore, doubling the firing rate quadruples axon volume. The concentration of mitochondria, an indicator of energy cost, tends to be constant with axon diameter; therefore, as volume quadruples, so does the energy supply (Perge et al., 2009, 2012). In summary, there is a *law of diminishing returns*: cost per bit, both in energy and space, rises steeply with bit rate (figure 3.6).

Three principles of neural design

The inescapable cost of sending any information and the disproportionate cost of sending at higher rates lead to three design principles: *send only what is needed*; *send at the lowest acceptable rate*; *minimize wire, that is, length and*

diameter of all neural processes. This last principle seems obvious, but it actually reflects a subtle point that arises from the constraint on rate.

Designs should reduce wire, of course, because wire uses space and energy. But wires also use *time* for transmission, and that is time lost to processing and action (Howarth et al., 2012). The constraint is particularly onerous for neural wires because they transmit more slowly than copper wire. Neural conduction velocity is 100 millionfold lower and, for biophysical reasons, faster conduction requires thicker wires (chapter 7). Thus saving time by sending at higher information rates (bits per second) and higher conduction velocities (meters per second) requires thicker axons, which, as noted, involves disproportionate costs in energy and space (Wen & Chklovskii, 2005). Thus, the only economical way to save time is to rigorously shorten wires. This principle shapes brain design across all scales, from an axon's branching and the microscopic design of local circuits, to the overall layout (chapter 13).

With these few principles we can now consider how the mammalian and fly brains are organized on a scale of about 1 mm and why. This macroorganization cannot explain the actual computations because those occur mostly on a finer scale. Nor do we claim that every feature represents the best of all possible designs. Others might work just as well—but they have not been tested. All we can say is that these three principles illuminate the layout of real brains—across a millionfold range of scale and half a billion years of evolution.